Until we are all equal

Participatory Climate Risk and Vulnerability Assessment for Gender Responsive Climate Adaptation and Preparedness Plans (GRAPP) in Kilifi County

Final Mnarani Ward Adaptation Plan

Consultant

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1 Chapter 1: Background information

1.1 Introduction

Plan International Kenya is implementing the Conservation and Sustainable Management of Marine and Coastal Ecosystems (COSME) Project in Kilifi and Kilifi Counties. The two counties are endowed with rich coastal and marine ecosystems ranging from terrestrial forests, mangroves, seagrass beds, coral reefs, marine species and open sea marine resources among others. These natural assets not only underpin livelihoods but also drive key economic sectors including agriculture, livestock, tourism, fisheries, and maritime trade. Despite their economic and ecological value, the 2021 country and county climate risk profiles highlight that these resources are increasingly under threat from climate change impacts such as drought, flooding, pest and diseases, environmental degradation, marine litter and dumping and strong winds. The consequences of these risks exacerbate underlying socio-economic vulnerabilities, with women, youth, and Persons Abled Differently (PAD) experiencing disproportionate impacts due to structural inequalities. Loss of income, disease outbreaks, displacement, reduced productivity and damage to infrastructure are already affecting community well-being and resilience in the region.

In response, the COSME project aims to enhance women-led, climate-resilient livelihoods through ecosystem-based approaches. Hence, to support project implementation requires the development of Gender Responsive Adaptation and Preparedness Plans (GRAPPs) that integrate climate risk diagnostics with locally appropriate adaptation pathways. The Mnarani Ward Climate Change Adaptation Plan (WAP) is therefore prepared as part of the GRAPPs outputs. The ward which measures 16.0 km^2, lies on the southern bank of Kilifi Creek, directly opposite Kilifi Town, and forms part of Kenya's scenic north coast along the Indian Ocean with a population of 11,253 (5,468 male & 5,785female) (KNBS, 2019), faces a high vulnerability to climate change impacts. The ward's experiences a warm, humid tropical climate typical of the coastal region, with bimodal rainfall and temperatures remaining relatively constant throughout the year while high humidity is largely influenced by the proximity of the Indian Ocean and Kilifi Creek. It's economy and livelihoods depend heavily on climate-sensitive sectors such as oceans and coastal zones, water resources, food systems, ecosystems and biodiversity which are susceptible to climatic changes (Figure 1). These form the backbone of local livelihoods including, agriculture, fishing, livestock keeping, tourism, wine tapping, and small scale businesses i. e fruit and nut vendors. However, prolonged droughts, erratic rainfall patterns, floods, and other climate-related stresses have already affected community well-being and local development. Climate change threatens to erode development gains and degrade critical natural resources upon which local livelihoods are dependent, making adaptation a critical priority for the ward.

Recurring droughts, pest and disease, outbreaks, periodic flooding, pests and diseases, environmental degradation, strong winds and coral bleaching have already affected community well-being and local development. Climate change threatens to erode hard-won development gains, making adaptation a critical priority for the ward. In response, Kenya's policy framework has evolved to support local climate action, from the National Climate Change Response Strategy and Action Plans to the Climate Change Act 2023, which emphasizes devolved climate governance and resilience building at county and community levels. This Ward Adaptation Plan aligns with national and county climate change policies by mainstreaming adaptation into local planning and empowering the community to implement priority resilience measures.

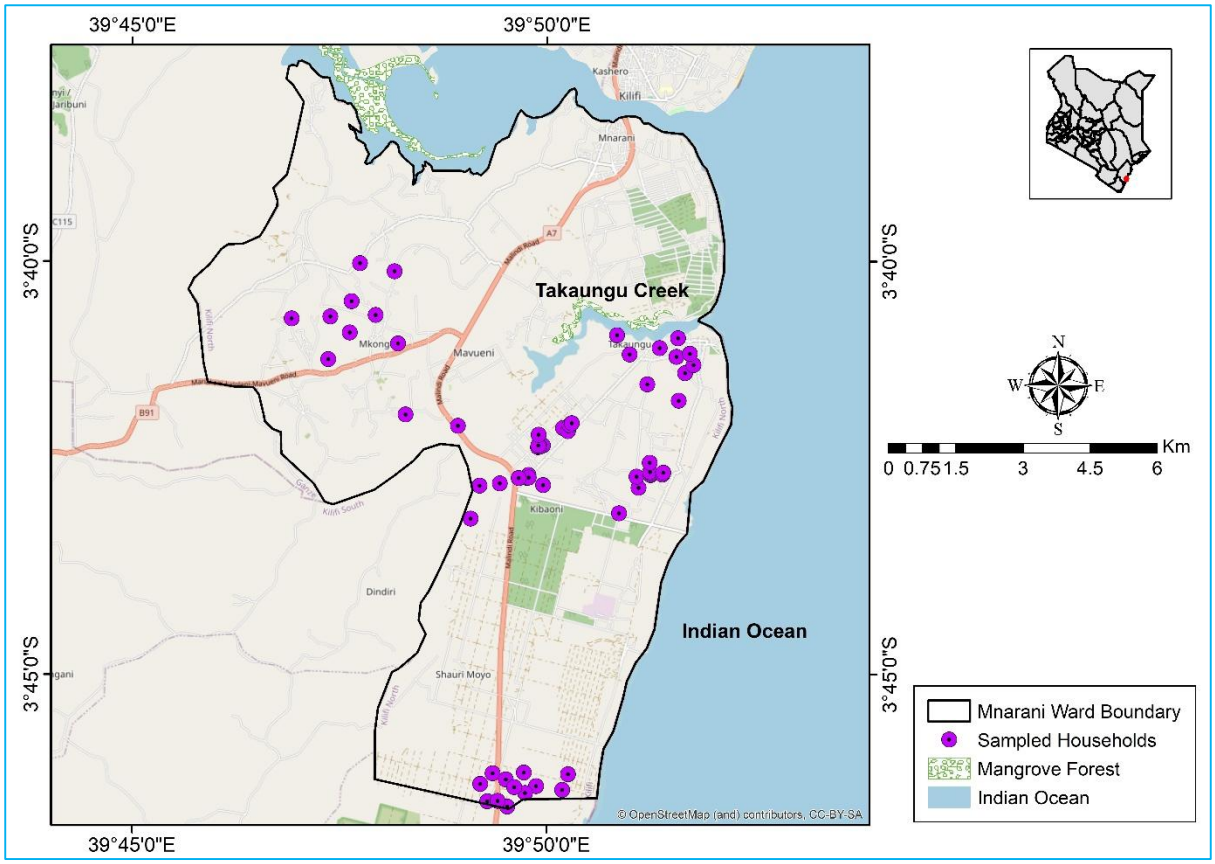

Figure 1: Location of Mnarani Ward and survey sites (Source: PCVA, 2025)

1.2 Plan development approach

The Ward Adaptation Plan for Mnarani was developed in a participatory, gender-responsive and livelihoods-based vulnerability framework that conceptualized climate risk as an outcome of the interaction between exposure, sensitivity and adaptive capacity (Figure 2). Within this analytical framing, vulnerability was understood as the degree of climate-induced damage experienced by socio-ecological systems, while resilience denoted the system's ability to maintain or rapidly regain functionality following climatic shocks. This approach acknowledged that climate risks in Mnarani, characterized predominantly by recurrent droughts, floods, pests and diseases, strong winds, coral bleaching and environmental degradation do not affect population groups uniformly. Women, men, youth, elderly persons and Persons Abled Differently (PADs) face differentiated risks due to disparities in livelihood assets, labor burdens, mobility, access to resources, intra-household power dynamics and decision-making autonomy.

The plan was therefore designed using a process that combined experiential community knowledge with quantitative datasets, spatial analysis, ecological diagnostics and climatic modelling. This integrated methodology enabled the plan to link climatic hazards to specific livelihood systems; rainfed agriculture, artisanal fishing, mangrove harvesting, small-scale trade and casual labour and to examine how these systems are shaped by gender norms, social vulnerabilities and institutional arrangements at ward level. The resulting insights provided a nuanced understanding of how climate extremes intersect with poverty, food insecurity, ecosystem degradation, weak infrastructure and limited institutional support, thereby shaping the context within which adaptation strategies must be developed.

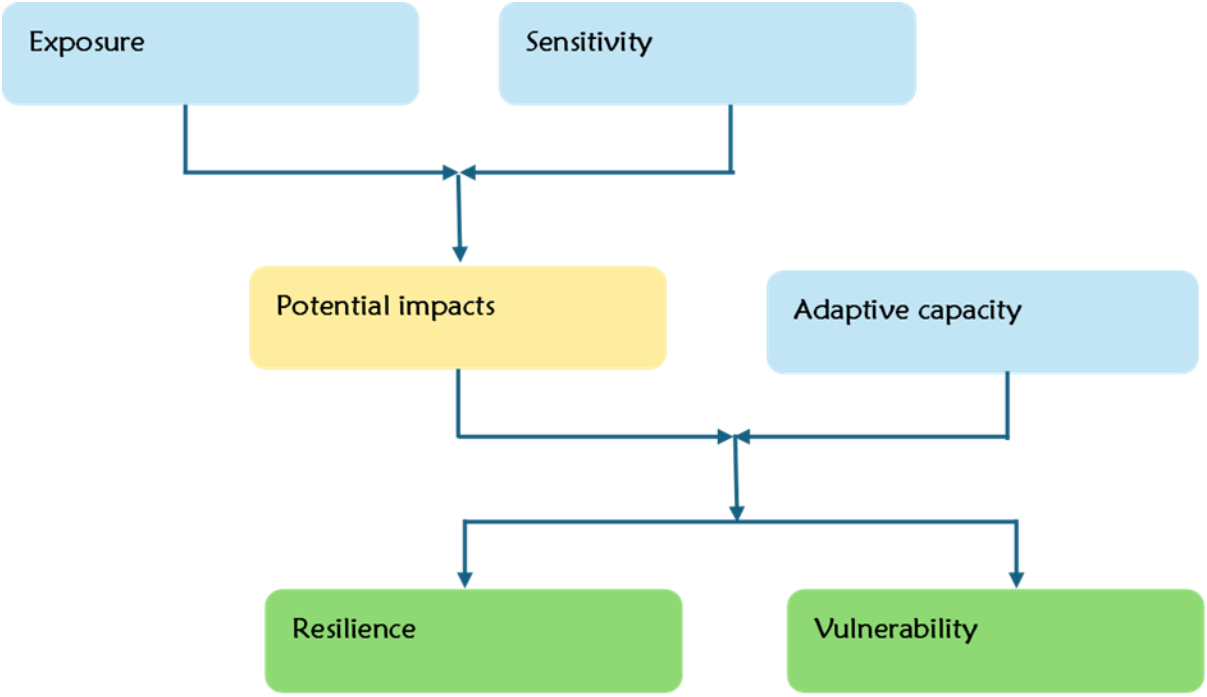

Figure 2: A model of factors that can be used to assess the vulnerability and resilience of systems (Source: Füssel and Klein, 2006, and Gauthier et al., 2014).

1.2.1 Inception Phase

The methodology was agreed upon through an inception phase that established a shared understanding among Plan International Kenya and the consultant on the objectives, scope and analytical expectations of the PCVA and WAP processes. An inception meeting held on 5th September 2025 brought together the COSME project teams to refine the methodological sequence, expected results, deliverable structure and timelines. During this phase, existing climate risk information for Kilifi County was reviewed, data gaps specific to Mnarani Ward were identified, and a sequenced work plan was developed outlining participatory diagnostics, household surveys, spatial and climate analysis, and subsequent synthesis and validation activities.

Following institutional alignment, data collection instruments were developed and technically refined to ensure that they were methodologically sound, gender-responsive and culturally appropriate. The instruments included ward-level FGD guides segregated for women, men and youth; participatory mapping and historical timeline guides; seasonal calendar and daily clock templates; household decision-making pile-sorting tools; Leaky Bucket diagrams; impact chain protocols; vulnerability matrix templates; institutional Venn diagram formats; and adaptation pathway worksheets. Each tool integrated prompts on gender norms, intersectional vulnerability, unpaid care workloads, labour shifts during climate stress and disability inclusion.

Stakeholder mapping during the inception phase identified relevant actors within Mnarani Ward, including the Ward Climate Change Planning Committee, chiefs and assistant chiefs, village elders, leaders of women and youth groups, PAD representatives and support networks, mangrove user groups, BMUs and community-based organizations. These actors facilitated community mobilization, venue preparation and dissemination of study information at ward level. The inception phase ensured that ward stakeholders were adequately oriented and positioned to meaningfully participate in the PCVA and adaptation planning process.

1.2.2 Participatory Climate Risk and Vulnerability Assessment (PCVA)

The Ward Adaptation Plan was informed by a comprehensive PCVA process implemented in six COSME target villages within Mnarani i.e., Midzimitsano, Mkwajuni, Takaungu Timboni, Vijiweni and Vuma. The assessment followed a structured six-step participatory sequence designed to diagnose climate vulnerabilities, identify underlying drivers of sensitivity, assess existing coping and adaptation strategies, and translate diagnostic findings into actionable adaptation options suitable for inclusion in the WAP. The process applied the full suite of ten PCVA tools, each deployed to generate gender- and age-disaggregated evidence capable of capturing complex interactions between climate hazards, livelihoods, natural resources and social systems.

1.2.2.1 Step 1: Scoping

During scoping, the geographical scope of the ward was delineated using GIS shapefiles provided by Plan International and Kilifi County Government. Within this boundary, the six COSME villages were purposively selected on the basis of ecological sensitivity, concentration of climate-sensitive livelihoods, recurrent exposure to droughts and floods, pests and diseases, susceptibility to strong winds, coral bleaching and environmental degradation. Village profiles were then developed to summarize demographic characteristics, livelihood systems, existing infrastructure, ecosystem conditions, climate-related impacts, past interventions and vulnerability hotspots. These profiles served as the baseline data against which PCVA tools were deployed.

1.2.2.2 Step 2: Establishment of Core Participatory Analysis Teams

A ward-level core participatory analysis team was established for Mnarani, drawing membership from Ward Climate Change Planning Committee (WCCPC) representatives, village elders, women leaders, youth leaders, PAD representatives and carers, NbS groups, GJJ groups, and other active CBOs and self-help groups. County technical officers from climate change, environment, forestry, agriculture, livestock and fisheries were engaged as resource persons.

The core team underwent a structured orientation on the PCVA framework, the ten PCVA tools, gender- and age-responsive facilitation, child safeguarding and ethical data collection. Training sessions used a combination of plenary presentations, group work and simulations of each PCVA tool, conducted primarily in Kiswahili to ensure comprehension. By the end of the training, the ward team had the capacity to co-facilitate hazard and resource mapping, historical timelines, seasonal calendars, daily clocks, household decision making and pile-sorting exercises, Leaky Bucket analyses, impact chain mapping, vulnerability matrix scoring, Venn diagrams and adaptation pathway development.

1.2.2.3 Step 3: Situation Analysis of Climate Vulnerability

The situation analysis integrated secondary datasets, primary field evidence and participatory analytics to generate a comprehensive picture of exposure, sensitivity and adaptive capacity across Mnarani Ward. This phase synthesised literature review, ward-level gender- and age-disaggregated Focus Group Discussions (FGDs), transect walks across the six COSME villages, and the structured application of PCVA Tools 1–7. Through triangulation, these methods produced a detailed, spatially nuanced and socially differentiated understanding of how climatic hazards interact with biophysical systems, livelihood structures and gendered social dynamics.

1.2.2.3.1 Literature Review

The literature review served as the technical backbone of the situation analysis, drawing from global, regional, national, county and historical ward-level evidence to contextualize observed vulnerabilities. At the global scale, the review considered findings from IPCC AR6 Working

Group I and II, particularly projections relevant to East African coastal systems, including accelerated warming rates, sea surface temperature rise, intensification of short-duration extreme rainfall, increased occurrence of coastal storms, saline intrusion risks and anticipated sea-level rise along the Western Indian Ocean. These findings were aligned with global adaptation and risk governance frameworks such as the Sendai Framework for Disaster Risk Reduction 2015–2030, the Paris Agreement adaptation cycle, and SDGs 2, 5, 6, 13 and 14, which collectively frame resilience building in coastal and marine socio-ecological systems.

At the regional scale, the review integrated IGAD Climate Prediction and Applications Centre (ICPAC) bulletins, the Africa Climate Policy Centre (ACPC) projections for Eastern Africa, and Western Indian Ocean Marine Science Association (WIOMSA) research on mangrove decline, fishery collapses and oceanographic shifts. This regional evidence positioned Mnarani within a broader coastal climate risk corridor characterised by rising temperatures, altered monsoon circulation, changing upwelling systems and degradation of nearshore ecosystems.

National climate policy and regulatory frameworks were examined comprehensively, including the Climate Change Act (2023), National Adaptation Plan (NAP 2015–2030), Kenya's Updated NDC (2023), the National Framework for Climate Services (NFCS), Kenya Climate-Smart Agriculture Strategy, National Disaster Management Policy, National Drought Early Warning System, and EMCA (2015) and sectoral regulations. These documents provided the lens for interpreting institutional mandates, priority hazards and adaptation responsibilities that cascade to county and ward levels.

At county scale, the review consolidated information from the Kilifi County Integrated Development Plan (CIDP III, 2023–2027), the Kilifi County Climate Change Act (2021), the County Climate Change Fund (CCCF) Regulations, the County Climate Change Action Plan (CCCAP 2023–2027), the County Spatial Plan, and sector strategies in agriculture, livestock, fisheries, water and coastal management. These county frameworks were triangulated with socio-economic and food security trends reported by the National Drought Management Authority (NDMA), historical rainfall data from the Kenya Meteorological Department (KMD), marine ecosystem studies from KMFRI, forestry and mangrove assessments by the Kenya Forest Service (KFS), fishery catch records from the State Department for Fisheries, and demographic data from the Kenya Population and Housing Census (2019).

The review also incorporated previous PCVAs conducted under COSME and other coastal adaptation programmes, which provided comparative vulnerability patterns, gender-responsive insights, and past lessons on nature-based solutions, community-based early warning systems, and local adaptation practices. This multilevel synthesis created a scientifically grounded baseline against which primary data from Mnarani could be interpreted.

1.2.2.3.2 Focus Group Discussions (FGDs)

Ward-level FGDs were conducted with segregated groups of women, men and youth from the six COSME villages; Midzimitsano, Mkwajuni, Takaungu Timboni, Vijiweni and Vuma, to elicit gendered and generational insights into climate trends, livelihood impacts and socio-ecological vulnerability (Figure 3). The discussions explored perceived changes in rainfall reliability, including delayed onset and inconsistent cessation of rains, prolonged intra-seasonal dry spells, unprecedented periods of elevated temperatures, shifts in wind patterns and increased intensity of episodic storms and coastal surges. Participants described the material impacts of these climatic variations on agricultural productivity, artisanal fisheries (including reduced catch per unit effort), mangrove health, livestock mortality and food security.

The FGDs also highlighted the deeply gendered nature of climate vulnerability. Women described increased walking distances and time burdens for water and fuel collection during droughts; heightened caregiving demands during disease outbreaks; increased exposure to GBV during water scarcity periods; and disruptions in women's informal trading activities. Youth reported loss of wage labour opportunities during climate shocks, stress-related migration, and lack of access to climate-resilient livelihood assets. Men described reduced fishing days caused by strong winds, damaged boats during storms and increased anxiety over household provisioning responsibilities. In addition, participants highlighted PAD-specific vulnerabilities, including mobility limitations during floods and exclusion from labour-intensive coping strategies.

These discussions produced climate calendars, gender-disaggregated vulnerability matrices and hazard trend narratives that deepened the temporal and social dimensionality of the vulnerability assessment.

Figure 3: FGDs in Mnarani Ward (Source: Field surveys, 2025)

1.2.2.3.3 Transect Walks

Transect walks undertaken across the six villages provided on-ground spatial verification of vulnerabilities identified in FGDs and documented in literature. These walks traversed rainfed agricultural plots, irrigated horticultural fields, mangrove stands and adjacent intertidal zones, fish landing sites, boreholes and shallow wells, public institutions (schools, health dispensaries), access roads, market areas, and known flood plains.

Structured observation sheets and GPS-enabled devices were used to capture geospatial attributes of key vulnerability hotspots. Evidence recorded included deterioration of feeder roads due to repeated flooding, heavily eroded gullies formed by runoff from bare catchments, siltation of wetlands, degraded mangrove stands exhibiting canopy thinning or sediment stress, and infrastructure damage such as washed-out culverts and submerged pathways.

These observations enhanced the physical realism of the vulnerability analysis and provided geo-referenced evidence that later strengthened hazard–resource maps and the identification of priority zones for ecological restoration and climate-resilient infrastructure

1.2.2.3.4 Tool 1: Hazard and Resource Mapping

The hazard and resource mapping tool enabled community members to visually represent their environmental and socio-economic landscape (Figure 4). Through participatory sketching, communities identified settlement clusters, farmlands, grazing areas, mangrove forests, rivers, markets, schools, health facilities, water points and religious centres. Hazard features; such as recurrent flood zones, degraded areas, storm-surge-prone coastal segments, areas and strong-winds corridors were marked and discussed. This mapping exercise generated spatially explicit knowledge that later informed adaptation targeting and zoning.

Figure 4: Hazard and resource mapping in Vuma Village, Mnarani Ward (Source: PCVA, 2025)

1.2.2.3.5 Tool 2: Historical Timelines

Historical timelines reconstructed multi-decadal climatic and socio-economic events including major droughts, destructive floods, storm events, pest invasions (armyworms and whiteflies), fishery fluctuations linked to monsoon variability and strong winds. Each event was discussed in terms of magnitude, social and economic impact, community responses and institutional support. Timelines revealed a pattern of intensifying climatic shocks, shortened recovery cycles and cumulative stress on natural resource systems.

1.2.2.3.6 Tool 3: Seasonal Calendars

Seasonal calendars detailed intra-annual cycles of rainfall, temperature, wind regimes, tidal patterns, agricultural operations, fishing seasons, mangrove harvesting, disease outbreaks and social events. Comparisons between historical and current seasonal behaviour revealed declining predictability of rainfall, increased occurrence of off-season rainfall, persistent dry spells and shifts in traditional fishing seasons. Calendars also captured periods of heightened GBV, school absenteeism, food scarcity and increased care burdens.

1.2.2.3.7 Tool 4: Daily Clock

The daily clock exercise revealed gendered labour allocation and how climate stress reconfigures time use. Under climate-normal conditions, tasks such as farming, childcare, water collection and marketing followed relatively stable patterns. Under climate stress, women and girls reported significant increases in unpaid care work and water collection time, men noted prolonged periods of income loss due to weather-inhibited fishing, and youth described increased domestic workload substitution and reduced learning hours. These insights demonstrated how climate shocks deepen time-poverty and reduce adaptive capacity.

1.2.2.3.8 Tool 5: Household Decision-Making Pile Sorting

Pile sorting exercises clarified intra-household power/authority structures under normal and climate-stress scenarios. Under normal conditions, decision-making on crop choices, livestock sales and minor household investments tended to be joint or gender-specific depending on cultural norms. Under climate stress, however, the decision space often shifted toward male dominance, particularly in decisions involving financial risk, fishing activity, asset disposal or climate-response investment, which in turn constrained women's adaptive agency and reinforced gender-based vulnerabilities.

1.2.2.3.9 Tool 6: Leaky Bucket

The Leaky Bucket tool illustrated how climate shocks disrupt household financial stability. Communities mapped income streams such as farming, fishing, petty trade, remittances and casual labour, alongside expenditure flows including food purchases, health costs, water, school fees, debt servicing and transport. Climate shocks were shown to reduce inflows drastically while simultaneously raising outflows through rising food prices, repair costs and medical expenses. This widening income–expenditure gap demonstrated the structural economic vulnerability of households.

1.2.2.3.10 Tool 7: Impact Chains

Impact chain analysis depicted the cascading consequences (direct and indirect impacts) of priority hazards. For instance, prolonged drought triggered crop losses, water scarcity, livestock mortality and reduced milk intake, which in turn contributed to household food insecurity, malnutrition risk among children, increased labour migration by men, rising care burdens for women, heightened GBV risk and reduced school attendance. Similarly, strong winds disrupted fishing, leading to economic losses, debt accumulation and psychological stress. This tool exposed multi-dimensional vulnerability pathways and supported the design of integrated adaptation strategies.

1.2.2.3.11 Step 4: Analysis of Causes of Vulnerability (Tool 8 – Vulnerability Matrix)

The analysis of underlying causes of vulnerability was undertaken through the application of the vulnerability matrix (Tool 8), which provided a structured means of examining how different climatic hazards intersect with livelihood assets and socio-ecological systems in Mnarani Ward. Hazards such as droughts, floods, pests and diseases, strong winds, coral bleaching and environmental degradation were arranged against critical assets including rain-fed farms, mangrove forests, artisanal fisheries, livestock, domestic water sources, housing structures, schools, health facilities, feeder roads and market infrastructure. Community members assigned severity ratings for each hazard–asset interaction, supported by qualitative reasoning and lived examples.

The cumulative scoring process produced a ranking that demonstrated, for example, the high susceptibility of mangrove areas and coral reef to combined effects of temperature stress, strong winds; the acute vulnerability of rain-fed agriculture to prolonged drought and unpredictable rainfall; and the differentiated impacts of floods on schools and feeder roads which disrupt

mobility, market access and educational continuity. This step bridged descriptive hazard analysis with deeper socio-economic and institutional determinants of vulnerability, forming the foundation for identifying priority adaptation pathways.

1.2.2.3.12 Step 5: Analysis of Community Action (Tool 9 – Venn Diagrams)

Analysis of existing coping and adaptation practices revealed a range of coping strategies and externally supported interventions. The Venn diagram exercise (Tool 9) enabled communities to map the institutional ecosystem influencing adaptation in the ward. Institutions such as county line departments, NGOs, community-based organisations, religious institutions, savings and credit groups, fisheries and forestry officers, COSME project structures, and private sector actors were placed on diagrams according to their perceived influence and accessibility. The spatial arrangement revealed a significant presence of institutions but varying degrees of functional support. While some organisations maintained close, supportive relationships, such as community groups collaborating with NbS and mangrove user groups, others operated at a distance, limiting their impact. This analysis also identified critical gaps, including insufficient extension services for fisheries, weak linkages to financial services, limited climate information dissemination, and inadequate institutional frameworks for systematic PAD inclusion. These findings provided essential insights for designing governance and institutional strengthening elements within the Ward Adaptation Plan.

1.2.2.3.13 Step 6: Drawing Action from Analysis (Tool 10 – Adaptation Pathways)

The final step in the participatory diagnostic process involved translating identified vulnerabilities, exposure pathways and institutional gaps into a coherent adaptation strategy through the Adaptation Pathways tool (Tool 10). Communities revisited impact chains and vulnerability dynamics to identify feasible response strategies across short-, medium- and long-term horizons. The exercise involved structured deliberations that considered gendered labour implications, the availability of financial and natural resources, household and community capacities, and alignment with ongoing COSME and county initiatives.

Resulting adaptation pathways articulated sequenced actions for key socio-ecological systems; agriculture, water resources, mangrove forests, coral reef and fisheries, community infrastructure, early warning systems and social protection mechanisms. This included immediate resilience-building measures such as rainwater harvesting, training and supporting women and youth on resilience design techniques, tree planting, and household-level livelihood diversification; medium-term actions such as installation of solar powered water pumps on existing wells, provision of farm inputs, supporting CFAs to implement their PFMP programs and strengthening local governance structures; and long-term interventions such as large-scale mangrove rehabilitation, drilling and solarizing boreholes, construction of water pans, institutionalization of community-based climate monitoring systems, and improvement of rural infrastructure. These pathways formed the backbone of the Mnarani Ward Adaptation Plan, linking diagnostic insights to strategic action.

1.2.3 Household Survey

A structured household survey was administered to 295 households in the study area to provide quantitative evidence complementing qualitative findings. Sampling was designed to include households across diverse livelihood portfolios and different vulnerability categories, including female-headed households, large households with high dependency ratios, households with PAD members, and those primarily reliant on climate-sensitive livelihoods such as rain-fed agriculture and artisanal fishing. Primary respondents were predominantly women (approximately 65 %), ensuring robust capture of intra-household dynamics, care burdens, access to services and lived experiences of climate stress.

The survey collected data on demographic structure, education, disability status, land tenure and resource access, livelihood contributions, income diversification, exposure to recent climatic shocks, losses and damages, coping strategies, food security indicators, institutional linkages and participation in social protection and savings groups. Data were collected digitally using KoboToolbox to ensure consistency and reduce transcription errors. These data provided a statistically grounded basis for validating PCVA findings and understanding patterns of vulnerability at household level across gender, age and livelihood categories.

1.2.4 Current Mangrove and Forest Health Assessment

To assess the condition of key coastal and terrestrial ecosystems supporting nature-based adaptation in Mnarani, a remote sensing analysis was conducted using Sentinel-2 imagery at 10-metre spatial resolution. Imagery underwent standard pre-processing steps including radiometric correction and cloud masking before the Normalised Difference Vegetation Index (NDVI) was computed using the formula.

$$\text{NDVI} = \frac{NIR - RED}{NIR + RED} \qquad \text{Equation 1}$$

Where RED is visible red reflectance, and NIR is near infrared reflectance.

NDVI values were categorised into low, moderate and high vegetation vigor classes. Low NDVI signatures corresponded to degraded or stressed vegetation, often reflecting factors such as saltwater intrusion, wood extraction pressure, sedimentation and hydrological alteration. High NDVI zones served as ecological benchmarks, indicating healthier vegetation stands. These spatial patterns were subsequently compared with PCVA hazard–resource maps and field observations to identify priority sites for ecological restoration, mangrove replanting, mangrove hydrological rehabilitation and sustainable forest management within the ward.

1.2.4.1 Current and Future Climate Projections

Climate projections relevant to Mnarani were developed using downscaled county-level analyses supplemented by remote sensing and historical observational datasets. Daily rainfall Climate Hazards Group InfraRed Precipitation with Station Data (CHIRPS), Actual Evapotranspiration (AET), Potential Evapotranspiration (PET), (from ERA5-Ag) and temperature variables (minimum, mean, maximum and Land Surface Temperature (LST) from Landsat and ERA5-Ag) were extracted through Google Earth Engine after spatial boundaries had been validated in QGIS. The resulting datasets were aggregated into annual totals or means and subsequently analysed using Python (Colab) employing pandas, numpy, scikit-learn and matplotlib.

Linear regression models were fitted to each climatic variable to estimate historical trends and project future changes to 2035 and 2045. Temperature projections incorporated Intergovernmental Panel on Climate Change (IPCC) AR6-aligned warming increments for East Africa's coastal belt, applying approximately +0.02°C per year for the moderate-emission scenario and +0.035°C per year for high-emission trajectories. Rainfall projections followed empirically derived trends, showing increasing variability, changes in seasonal distribution and heightened likelihood of intense episodic rainfall events. These climatic insights were interpreted in relation to Mnarani hazard profile, reinforcing the need for enhanced water security, climate-resilient agricultural practices, improved coastal and marine ecosystem management, stronger infrastructure planning and expanded early warning systems.

1.2.5 Data Analysis and Reporting

Quantitative data from the household surveys were cleaned, coded and analyzed using Microsoft Excel and Python. Analysis focused on frequencies, proportions, cross-tabulations and simple correlations across gender, age group, livelihood type and vulnerability status. These results provided empirical confirmation of patterns emerging from qualitative insights.

Qualitative data from FGDs, PCVA exercises and transect walks were transcribed and thematically coded to organise evidence under key domains such as hazard exposure, livelihood impacts, gender dynamics, coping strategies, institutional barriers and opportunities. Outputs from PCVA tools; maps, timelines, calendars, clocks, matrices and impact chains, were digitized and integrated with NDVI layers and climate trend visualizations to produce synthesized analytical outputs. Triangulation across all sources ensured internal consistency and improved the robustness of the final interpretation. Draft findings were compiled for review by Plan International Kenya in line with expected deliverables.

1.2.6 Validation

The draft findings were presented in county-level validation workshops held in Kilifi and Kilifi on 20th and 21st November 2025 respectively. These sessions provided a platform for county technical departments, CCPC/WCCPC members, civil society, COSME partners and community representatives to review the PCVA results and refine the draft adaptation strategies. At ward level, stakeholders from Mnarani reviewed hazard rankings, adaptation priorities and institutional roles, and provided further refinement on feasibility, gender inclusion, sequencing and resource implications. Inputs from these processes were incorporated into the final Ward Adaptation Plan to ensure scientific accuracy, community legitimacy and institutional alignment.

1.2.7 Linkages to other planning processes

At both the National, County and Ward Levels, several other planning processes exist to which the WAP is linked. These include the National Adaptation Plan 2015-2030, the Kilifi County Adaptation Plan 2023-2032 and Mnarani Ward level Participatory Climate Risk Vulnerability Assessment (PCVA) conducted in 2023. The NAP is Kenya's first plan on adaptation and demonstrates the commitment to operationalize the National Climate Change Action Plan by mainstreaming adaptation across all sectors in the national planning, budgeting and implementation processes. The mainstreaming approach recognizes that climate change is a cross-cutting sustainable development issues with economic, social and environmental impacts. Similarly, the CAP aligns with the NAP in terms of content and reporting focusing on Kilifi County circumstances, climate hazards and adaptation actions which are largely derived from PCVAs for the 35 wards. The WAP adaptation plan is prepared considering both the NAP and CAP and aligning adaptation actions accordingly as informed by the village focused PCVAs.

2 Chapter 2: Coordination of Adaptation Actions

2.1 Introduction

Climate change adaptation requires a coherent and inclusive governance framework that aligns actions across sectors, levels of government, and communities. In Mnarani Ward, where climate risks affect water access, agriculture, health, infrastructure, livestock, and coastal ecosystems, coordination ensures that diverse actors i.e government institutions, civil society, Non-Governmental Organization (NGOs), funding partners, research bodies, and local communities work together toward shared resilience goals. This chapter focuses on the governance architecture needed to translate the Ward Adaptation Plan into action by linking local priorities with county programming and national direction, while ensuring timely access to finance, technical support, and climate information. The coordination approach combines horizontal collaboration among village institutions, civil society, private actors, and technical officers with vertical integration that embeds ward priorities into county plans and Kenya's National Adaptation Plan (2015–2030). By harmonizing strategies, clarifying roles, and promoting gender-responsive, locally led initiatives, this system minimizes duplication, strengthens accountability, and transforms adaptation commitments into measurable, sustainable resilience outcomes.

2.2 Current Institutional Arrangements

Coordination presently operates across three interlinked tiers. Kenya's national framework for climate change adaptation and resilience is anchored in the Climate Change Act (2023), which provides the legal foundation for coordinated climate action across all levels of governance. The National Climate Change Council and the Climate Change Directorate provide policy direction, coordination, and technical backstopping to ensure that adaptation measures are integrated into national and county development processes. Operational guidance is delivered through the Kenya National Adaptation Plan (NAP) 2015–2030 and the National Climate Change Action Plan (NCCAP) 2023–2027, which outlines priority actions to enhance adaptive capacity and reduce vulnerability across key sectors. Specialized agencies reinforce this institutional architecture: the Kenya Meteorological Department provides forecasts and impact-based early warnings; the National Drought Management Authority (NDMA) leads drought monitoring and response; the Coast Development Authority (CDA) coordinates integrated coastal zone development, conservation, management and sustainable use of coastal and marine resources; and the National Environment Management Authority (NEMA) enforces environmental safeguards for adaptation investments. These agencies coordinate with county structures and engage directly with wards, ensuring that national policies are translated into locally relevant, gender-responsive, and climate-resilient actions.

At county level, Kilifi County Government acts as the bridge between national strategy and local action. The Kilifi County Climate Change Act (2021) provides a legal framework for coordinated climate action within the county, mandating the County Government to establish institutional structures, mobilize and allocate resources for climate adaptation and mitigation, and ensure that vulnerable groups benefit equitably from climate financing. The Act's implementation is operationalized through the Kilifi County Climate Change Action Plan (2023–2027), which outlines strategic interventions aligned with national and county priorities to enhance resilience, promote sustainable livelihoods, and guide sectoral climate responses. Additionally, the County Integrated Development Plan (CIDP) 2023–2027, integrates climate change considerations into broader development planning, ensuring that climate actions are mainstreamed into the county's long-term socio-economic agenda and across various departments (Environment and Climate Change; Infrastructure and Transport; Agriculture and Blue Economy; Water and Sanitation; Health and Tourism) to integrate resilience criteria into annual work plans. Together, these frameworks position the county in climate governance, fostering inclusive, well-coordinated, and sustainable climate responses. County coordination

platforms such as the County Climate Change Steering and Planning Committees oversee the integration of climate adaptation and mitigation across sectors in Kilifi. These structures, supported by the Directorate of Climate Change, coordinate multi-stakeholder engagement, resource mobilization, and the implementation of the County Climate Change Action Plan. Moreover, partnerships with non-governmental organizations, development partners, and private sector actors are convened through county-led forums, promoting alignment of external climate initiatives with county priorities and public investments, thereby enhancing coordination, transparency, and impact.

At ward and community level, implementation energy is concentrated. Ward Climate Change Committees (WCCCs) serve as the primary coordination platforms for community-based climate action in Kilifi County. These committees are responsible for mobilizing local stakeholders to identify, prioritize, and implement climate-resilient projects that address specific vulnerabilities within their areas. Comprising representatives from community groups, youth, women, and local administration, the WCCCs ensure that climate initiatives are inclusive and grounded in local realities. They work closely with the County Climate Change Directorate to channel community priorities into county planning and budgeting processes. Through these committees, wards have become active spaces for participatory climate governance, fostering local ownership, enhancing accountability, and strengthening the link between grassroots adaptation efforts and county-level policy implementation.

2.3 Relevant Policies

The coordination framework for climate adaptation in Mnarani Ward is supported by a strong policy environment. At the national level National Adaptation Plan (2015–2030), National Climate Change Action Plan (NCCAP 2023–2027) and Climate Change Act (2023) provide the overarching structure for climate governance, emphasizing locally led adaptation and devolved financing. The National Adaptation Plan (NAP 2015–2030) identifies priority sectors relevant to Mnarani, including climate-smart agriculture, livestock development, water resource management, tourism and fisheries value chain enhancement, ecosystem restoration, disaster preparedness, gender, vulnerable groups and youth inclusivity and human health resilience. Programs such as the Conservation and Sustainable Management of Coastal and Marine Ecosystems (COSME Project) and Financing Locally Led Climate Action (FLLoCA) channel grants to support ward-level initiatives.

At the county level, the Kilifi County Climate Change Act (2021) and County Climate Change Action Plan, (2023–2027) provides an enabling framework for ward proposals to access the County Climate Change Fund (CCCF)as well as framework for mainstreaming adaptation into sectoral plans, including but not limited to Environment and Climate Change, Infrastructure and Transport, Agriculture and Blue Economy, Water and Sanitation, Health, and Tourism. Sector-specific policies further reinforce this framework: the Water Act, 2016 and National Water Policy, 2019 embed community-based water governance and integrated water resource management; Agricultural Sector Transformation and Growth Strategy, 2019 promotes diversified, climate-resilient production and value addition; National Disaster Risk Management Policy, 2021 strengthens anticipatory action and local preparedness while the National Gender and Climate Change Action Plan (NGCCAP) 2025-2027 ensure equitable participation and benefit sharing in climate initiatives. Together, these instruments establish the principles, standards, financing mechanisms, and safeguards that the Mnarani coordination system adopts, thereby ensuring consistency with the NAP while tailoring execution to place-based realities.

2.4 Proposed Coordination Arrangements

The proposed coordination framework for Mnarani Ward strengthens existing systems while introducing clear linkages, communication channels, and accountability mechanisms to support

coherent and inclusive implementation of adaptation actions. The arrangement mirrors national expectations for devolved climate governance under the Climate Change Act (2023) and the Kilifi County Climate Change Act (2021), ensuring alignment between community priorities and county-level planning, financing, and reporting processes.

At the ward-county interface, a designated County Climate Change Officer from the County Climate Change Unit (CCCU) will serve as the technical liaison between the WCCPC and the county departments. The officer will attend WCCPC meetings, share updates on ongoing county programs, and support integration of ward priorities into sectoral plans and the County Climate Change Fund (CCCF) proposal pipeline. The WCCPC, in collaboration with the Ward Administrator, will prepare quarterly progress reports summarizing implementation progress, budget utilization, gender-disaggregated data, challenges, and lessons learned. These reports will be submitted to the CCCU to ensure upward accountability and inform county planning and resource allocation.

Within the ward, the WCCPC will act as the central coordination and decision-making platform. It will convene a biannual Mnarani Climate Stakeholder Forum, bringing together all relevant actors including county departments, national agencies, NGOs, CFAs, WRUAs, self-help groups, private entities, and academic and research institutions. The forum will serve as a space to map ongoing interventions, harmonize technical standards, identify overlaps or gaps, and develop joint action plans. It will also facilitate knowledge sharing, mobilize resources, and enhance coordination between local and county-level actors.

To strengthen operational efficiency, the WCCPC will establish task-specific subcommittees responsible for thematic areas such as water management, forest and ecosystem restoration, climate-smart agriculture, and disaster preparedness. Each subcommittee will include representatives from relevant community groups, technical officers, and development partners, ensuring that actions are context-specific, technically sound, and gender-responsive. For instance, conservation partners such as Kenya Forest Service and World Bank to rehabilitate degraded forest areas, promote tree planting, and strengthen community forest governance structures as well as sustain sustainable fisheries livelihoods respectively

Given Mnarani vulnerability to recurrent droughts, pests environmental degradation, strong winds and floods, climate coordination will be closely linked with disaster risk management structures. The WCCPC will collaborate with the Ward Disaster Management Committee to ensure early warning systems, community awareness, and anticipatory actions are integrated into regular adaptation planning. Seasonal forecasts from the Kenya Meteorological Department (KMD) and early warning information from the NDMA will be disseminated through village communication tree i.e community baraza's and where feasible through means such as local radio station and mobile alerts to enhance preparedness and response.

Finally, an annual review and reflection meeting, co-convened by the CCCU and the WCCPC, will assess progress against the NAP indicators and the County Climate Change Action Plan (2023–2027) targets. This platform will document best practices, challenges, and emerging lessons to improve coordination, refine institutional arrangements, and strengthen adaptive management at the ward level.

2.5 Implementation Roles

Effective implementation of the Ward Adaptation Plan requires clear role delineation among all stakeholders to ensure that adaptation measures are executed efficiently, inclusively, and in line with Kenya's National Adaptation Plan (2015–2030). Table 1 below outlines the mandates, decision-making authority, and accountability mechanisms of the

key institutions and actors engaged in rolling out adaptation interventions in Mnarani Ward. The WAP explicitly mainstreams Gender Equality and Inclusion (GEI) across all stages of planning and implementation and establishes strong coordination linkages with health, gender, and protection actors to guarantee continuity of essential Sexual and Reproductive Health and Rights (SRHR) and GBV prevention and response services during climate shocks.

Table 1: Institutional Roles in Implementation of Adaptation Actions

Institution/Actor	Key Roles and Responsibilities
Ward Climate Change Planning Committee (WCCPC)	Lead planning, coordination, and prioritization of ward-level adaptation actions; mobilize community participation; ensure gender and social inclusion in project design and implementation.
Kilifi County Government Departments	Provide technical oversight, finance, and sectoral expertise (Climate, Water, Agriculture, Livestock, Fisheries, Roads, Health, Environment). Integrate ward priorities into departmental annual work and climate change plans, programs and budgets.
National Drought Management Authority (NDMA)	Lead drought monitoring, early warning, and response; integrate Mnarani into national drought early warning systems and coordinate emergency relief when thresholds are reached.
Coast Development Authority (CDA)	Coastal zone development, conservation, management and sustainable use of coastal and marine resources
Kenya Climate Innovation Centre (KCIC)	Supports climate-smart innovations, private-sector and agribusiness adaptation solutions
Kenya Meteorological Department (KMD)	Provide localized weather and climate forecasts, early warnings, and advisories for decision-making in agriculture, fisheries, and disaster preparedness.
Kenya Fisheries Services (KeFS)	Sustainable fisheries for coastal communities, diversification of livelihoods, and protection of marine ecosystems
Local community	Management, protection and restoration of Kayas
Beach Management Units	Marine, fisheries and coastal ecosystem protection, mangrove conservation
National Environment Management Authority (NEMA)	Review and approve Environmental Impact Assessments (EIAs) for adaptation projects; ensure adherence to environmental safeguards.
Development Partners (UNDP, ADB, USAID, UKAID, GCF, World Bank, EU, FAO, IFAD)	Provide technical assistance, capacity building, and climate finance to support implementation of local adaptation priorities; promote mainstreaming of climate resilience in development investments and infrastructure.
Non-Governmental Organizations (NGOs) and CSOs (Plan International, Redcross, World Vision Kenya)	Support implementation of community-based adaptation projects; promote gender-responsive climate resilience initiatives; facilitate training and awareness creation on nature-based solutions (NbS) and livelihood diversification.
Conservation groups and associations	Marine ecosystem restoration, coral reef protection
Community Forest Associations, (CFA), Community-Based	Implement localized adaptation initiatives such as mangrove restoration, water harvesting, and livelihood diversification; mobilize local labour and

Institution/Actor	Key Roles and Responsibilities
Organizations (CBOs) and Self-Help Groups	indigenous knowledge; ensure equitable participation in project benefits.
Private Sector and Cooperatives	Promote investment in climate-smart agriculture, renewable energy, and insurance products; integrate environmental sustainability in business operations; collaborate in corporate social responsibility (CSR) initiatives that support community resilience.
Academic and Research Institutions (KALRO, KMFRI, KEMFRI, Universities)	Generate and disseminate climate and ecological data; support participatory research and innovation for adaptation technologies; build capacity of local institutions on agriculture, livestock, ecosystem monitoring and early warning systems.

3 Chapter 3: Situation analysis of climate vulnerability

3.1 Introduction

Over the past years, Mnarani Ward has been exposed to a growing range of interlinked climatic hazards whose severity and persistence have progressively increased. The Participatory Climate Vulnerability and Capacity Assessment (PCVA, 2025) highlights recurring droughts, floods, pest and disease outbreaks, environmental degradation, strong winds and coral bleaching as the major stressors affecting local communities. These challenges have altered the functioning of ecosystems and disrupted the stability of livelihoods across the settlements of Vuma, Vijiweni, Takaungu, Mkwajuni, Timboni and Midzi Mitsano areas. The assessment applied an integrated methodology combining indigenous knowledge systems, historical trend analysis, and scientific projections to determine the frequency, magnitude, and spatial distribution of each hazard. Vulnerability was analyzed across six representative villages (Vuma, Vijiweni, Takaungu, Mkwajuni, Timboni and Midzi Mitsano) focusing on two key parameters: exposure to climatic stressors and adaptive capacity at both household and institutional levels.

Findings show that women, the elderly, widows, persons abled differently (PAD), and low-income families face the greatest risk. Their exposure was linked to heavy reliance on natural resources, limited financial and productive means, and the additional burden of unpaid care and household work during climate crises. The distribution of hazards across the ward reveals clear spatial patterns. Prolonged droughts are most common in Takaungu, Majajani, Kidundu, and Mnarani Town, where water scarcity, repeated crop failure, land degradation, and livestock losses are widespread. Flooding tends to occur in Mbogolo, Benesi, Mwakuhenga, Mnarani Town, and Majajani, often damaging roads, homes, and other infrastructure while disrupting livelihoods. Pests and diseases effects span across the ward, which have become increasingly severe, driven by rising temperatures, prolonged dry spells, pervasive poverty, and limited resources that collectively weaken community resilience across the ward. Environmental degradation, particularly visible in Vuma, Timboni and Takaungu villages was attributed to deforestation and expanding industrial and economic expansion, causing loss of fertile soils and increasing vulnerability to further climatic shocks. Moreover, the ward is experiencing emerging climatic hazards including strong winds and coral bleaching in Vuma and Timboni villages respectively, which have intensified recently attributed to both natural climatic variability and human-induced activities. Understanding these spatial and demographic dynamics provides a crucial evidence base for prioritizing adaptation interventions aimed at minimizing exposure, reducing vulnerability, and enhancing the long-term resilience of communities and ecosystems in Mnarani Ward.

3.2 Current climate trends

The ward experiences a tropical climate characterized by warm temperatures and marked inter-annual variability in rainfall. Based on baseline observations (1981–2010) and recent trends, annual rainfall averages vary with peaks recorded at approximately 1,600 mm, with year-to-year totals ranging between 1,200 mm and 1,700 mm (Figure 5). During the same period, Potential Evapotranspiration (PET) remains relatively stable at around 1,150 mm, while Actual Evapotranspiration (AET) averages approximately 1,650 mm annually. The relatively higher AET values in comparison to rainfall indicate that much of the available moisture is continuously cycled back to the atmosphere, driven by vegetation cover and prevailing temperatures. Although rainfall shows a slight increasing trend, the persistent evapotranspiration rates suggest that periods of water stress may still occur during drier months, particularly when seasonal rainfall fails to meet evaporative demand.

Temperature analysis under the same scenario (Figure 6) shows that mean air temperature increases from approximately 25.8 °C to 26.7 °C, while land surface temperature (LST) rises from about 34.4 °C to 34.8 °C. This warming trend may be attributed to factors such as

increased greenhouse gas concentrations, reduced vegetative cover, and growing urbanization, which enhance surface heat retention. Additionally, changes in local albedo and declining canopy shading likely contribute to higher surface temperatures. These conditions collectively promote stronger evapotranspiration rates and soil moisture depletion, potentially intensifying heat stress and altering local microclimates.

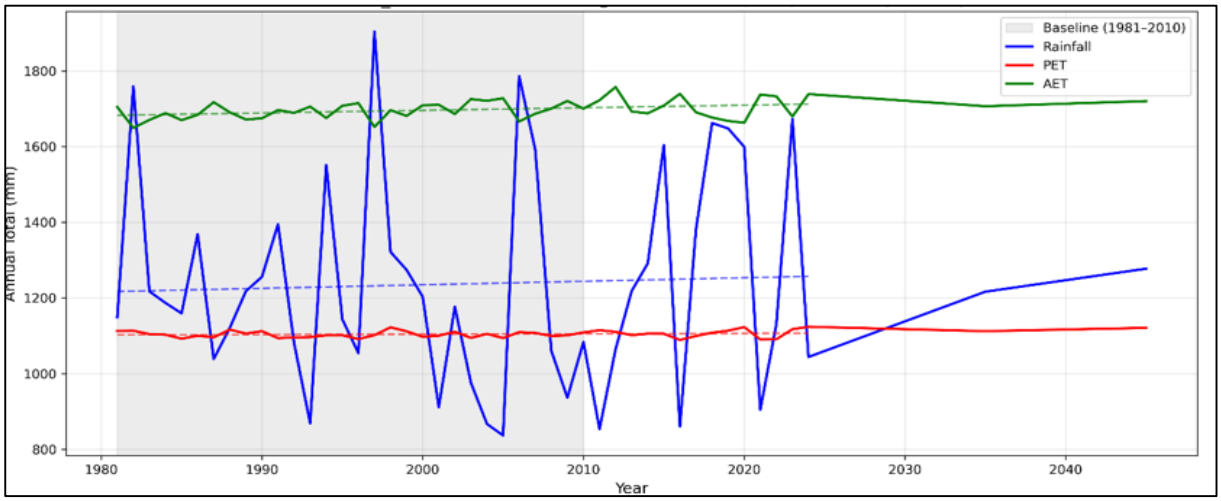

Figure 5: SSP2 4.5 Rainfall, AET and PET Projections. (Source: Chirps database and ERA5-Ag, 2022))

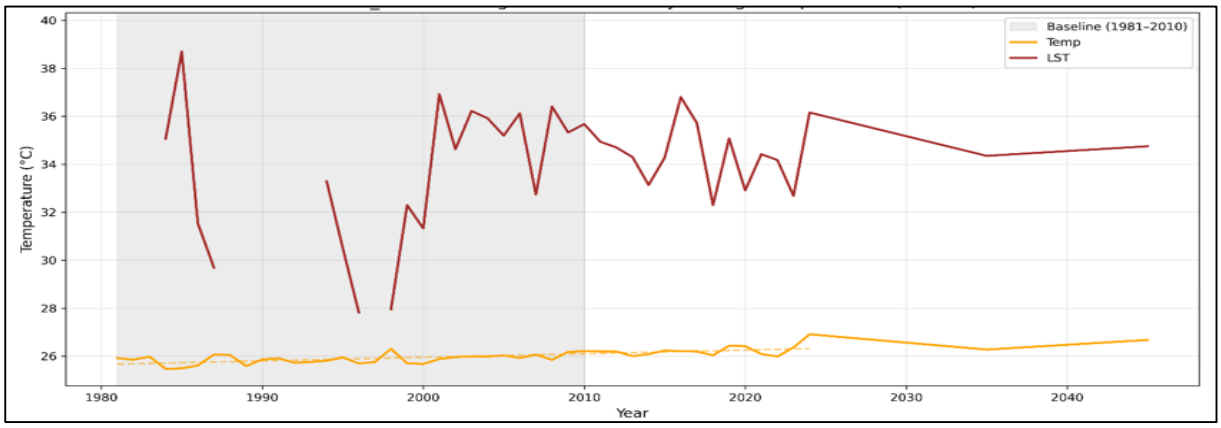

Figure 6: SSP2 4.5 Mean Temperature and Land Surface Temperature Projections. (Source: Chirps database and ERA5-Ag, 2022)

3.3 Future climate trends

Projected climate conditions under the high-emission scenario (SSP5–8.5) indicate that rainfall in Mnarani Ward is expected to increase from 1,253.0 to 1,338.2 mm, while potential evapotranspiration (PET) rises slightly from 1,123.2 to 1,136.4 mm, and actual evapotranspiration (AET) from 1,723.8 to 1,744.2 mm. The stronger rainfall signal points to generally wetter conditions; however, the concurrent increase in PET suggests heightened evaporative losses that may offset some of the anticipated water gains (Figure 7). The increase in rainfall suggests that Mnarani Ward will likely experience wetter conditions in the future. Although rainfall is expected to increase, higher temperatures will raise evaporative demand, limiting water availability and offsetting potential gains in the net water balance.

Projected temperature patterns under the high-emission scenario (SSP5–8.5) reinforce the ongoing warming trajectory in Mnarani Ward. The mean air temperature is projected to rise from 26.5 °C to 26.9 °C, accompanied by an increase in land surface temperature (LST) from 34.6 °C to 35.0 °C. This consistent warming trend reflects enhanced surface energy fluxes and

an elevated probability of heat extremes toward 2045 (Figure 8). The projected rise in mean temperature and LST reflects intensified surface heating driven by higher greenhouse gas concentrations. This warming increases the likelihood of heat extremes and thermal stress in Mnarani Ward toward 2045.

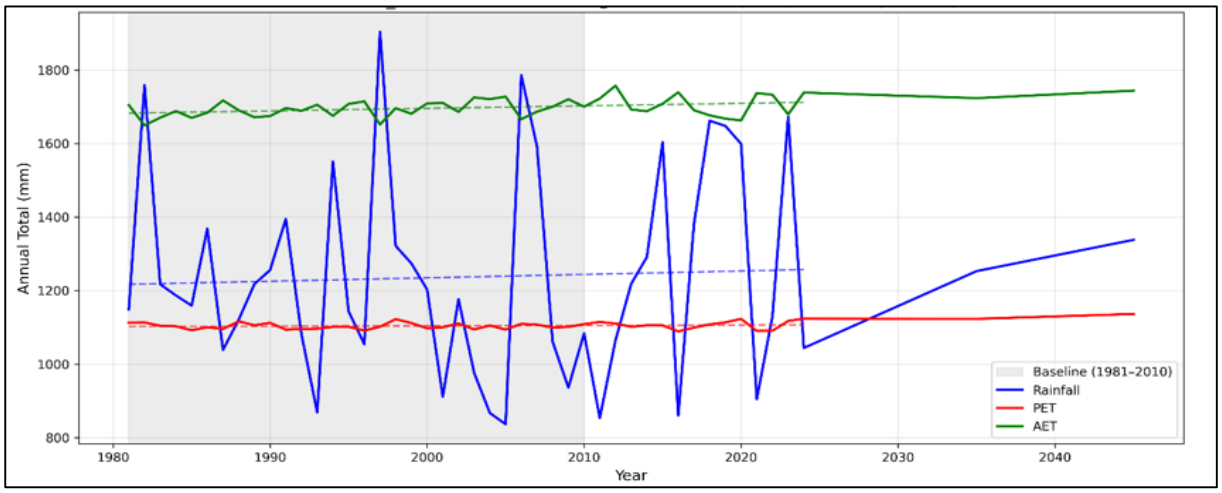

Figure 7: SSP5 8.5 Rainfall, AET and PET Projections. (Source: Chirps database and ERA5-Ag, 2022)

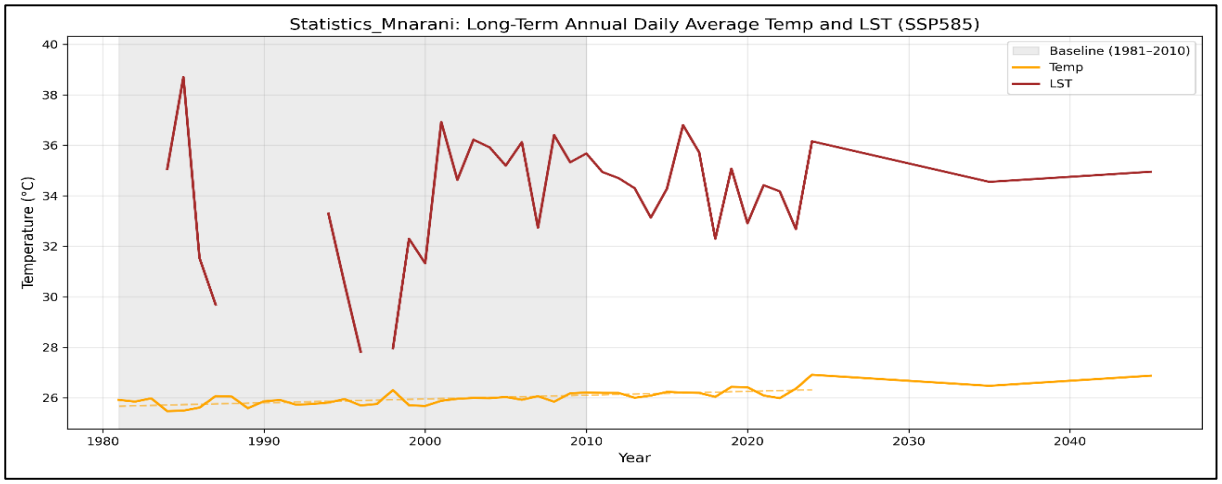

Figure 8: SSP5 8.5 Mean Temperature and Land Surface Temperature Projections. (Source: Chirps database and ERA5-Ag, 2022)

3.4 Climate hazards and vulnerability

3.4.1 Drought

Drought remains the most persistent and economically significant climate hazard in Mnarani Ward. It exerts sustained pressure on terrestrial and coastal ecosystems, disrupting livelihoods and reducing agricultural productivity. Historical time (1930–2025) indicate that major drought events occur approximately once every six years (Figure 7). Notable severe episodes were recorded in 1960 (*Nzala ya ndugu si mtu*), 1978 (*changilo*), 1980, 1982, 1997– 1998 and 2012 –2015. The analysis shows a clear upward trend in the frequency, duration, and intensity of droughts over time. The increasing severity of drought conditions is associated with a decline in rainfall reliability, delayed onset and early cessation of the rainy seasons, and higher evapotranspiration rates. These changes have collectively shortened growing seasons, increased crop failure rates, and intensified food insecurity across the ward.

Since the 1930s, vulnerability levels have varied in magnitude, with scores indicating that Midzi Mitsano (54), Timboni (49), Vuma (48) and Takaungu (46) exhibiting high exposure levels

which was attributed to reliance on agriculture activities, proximity to water resources, This heightened vulnerability is largely attributed to dependence on rain-fed agricultural activities, presence of water resources (rivers and streams), and proximity to mangrove and forest ecosystems (private and natural) that are increasingly affected by recurrent drought conditions. In contrast, Mkwajuni (35) and Vijiweni (26) record relatively lower exposure levels, primarily due to livelihood diversification and the availability of complementary water resources such as county-managed water tanks, boreholes, private wells as well as reticulated supply (Figure 8).

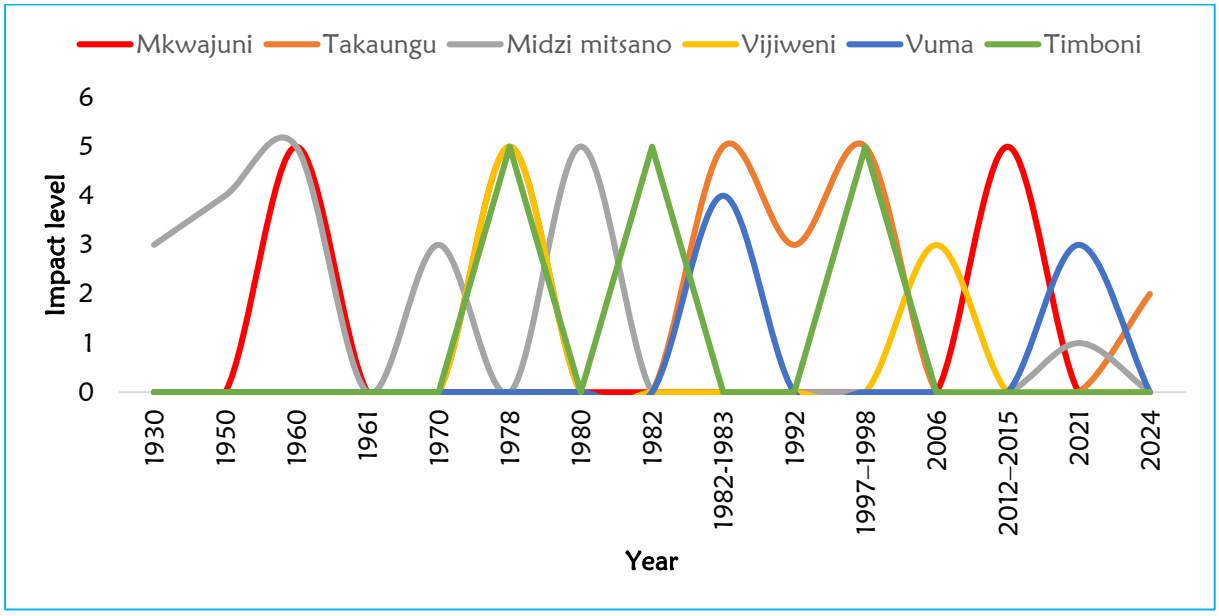

Figure 9: Drought prevalence historical trend in Mnarani (Source: PCVA, 2025)

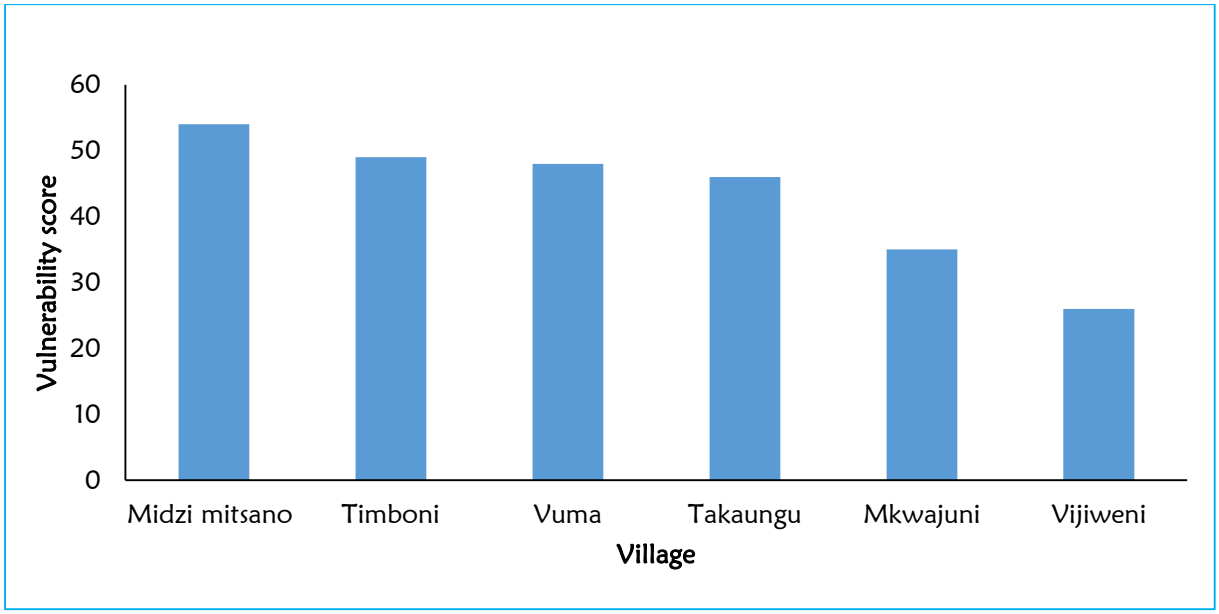

Figure 10: Drought vulnerability score in Mnarani Ward (Source: PCVA,2025)

Mangroves in Takaungu Creek display marked seasonal variability in vegetation health. During the dry season, NDVI values range from 0.24 to 0.49, indicating generally moderate to healthy vegetation with some areas experiencing mild stress due to limited freshwater inflow and elevated salinity. In the wet season, NDVI values vary widely from 0.06 to 0.51, reflecting both improved growth in some areas and significant stress in others. The very low values observed in parts of the creek during the wet season may be attributed to water saturation effects,

flooding, or sediment deposition, which can reduce oxygen availability to roots and hinder regeneration. These patterns suggest that Takaungu Creek mangroves are highly sensitive to hydrological changes, with localized wet-season stress potentially counteracting the benefits of increased rainfall (Figure 9).

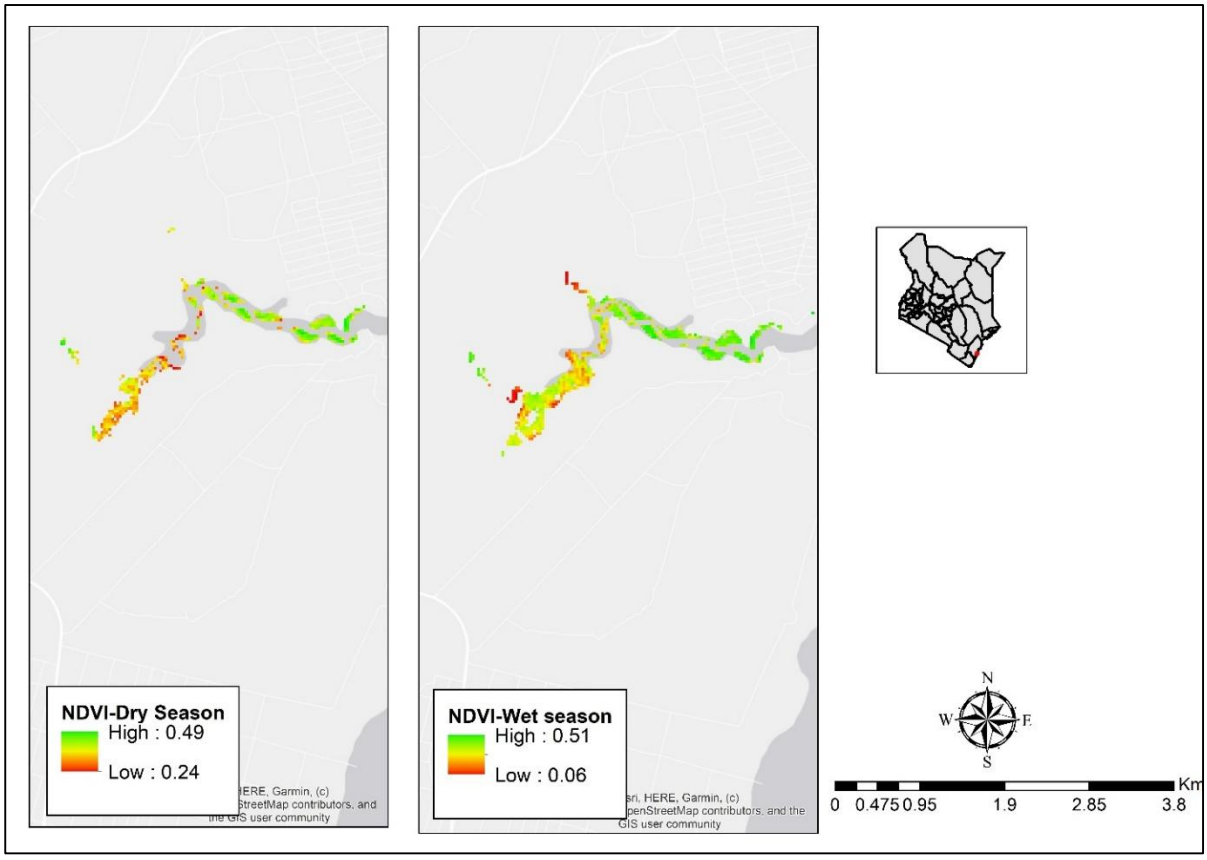

Figure 11: Takaungu Creek mangrove forest NDVI values across dry and wet seasons (Source: Source: Landsat 8/9 OLI and TIRS, 2024)

These variations reflect the combined influence of localized climatic variability, ecosystem degradation, and unequal adaptive capacities among communities. Spatially, drought hotspot zones are concentrated around riverine systems such as Taratibu, Kombeni, and Mavueni, and at key water pan sites including Mbogolo, Mahara, Benes, Kwa Kisiwa, Ngondo, and Mabirikani. Additional high-exposure zones are observed around mangrove areas and the Kambi ya Shetani Forest ecosystem, where declining vegetation cover and water stress have intensified environmental vulnerability.

At the community level, drought exposure is not experienced equally. Women, who are primarily responsible for water collection, household food security, and caregiving, face disproportionate burdens during dry periods. The long distances traveled to fetch water from depleted sources, often in unsafe conditions, limit women's time for income-generating activities and expose them to physical and psychosocial risks. Reduced agricultural yields also affect women's ability to provide food for their families, exacerbating household poverty and malnutrition. Moreover, their limited access to land ownership and climate information further constrains their capacity to cope with and recover from drought impacts. This intersection of environmental stress and gender inequality deepens the overall exposure of communities to drought. As natural resources degrade, women's workloads increase while their livelihood options shrink, perpetuating cycles of vulnerability. Addressing these challenges requires integrated adaptation measures that strengthen water storage and access, promote gender-

responsive resource management, and empower women through targeted capacity-building, livelihood diversification, and leadership in conservation initiatives around mangroves and Kayas.

3.4.1.1 Exposure to Drought

Exposure to drought in Mnarani Ward is extensive and interconnected, affecting both ecological systems and community livelihoods across all major villages. Drought events, which occur approximately once every six years, cause widespread depletion of surface and groundwater sources, resulting in acute water shortages for domestic use, livestock, and irrigation. Farmlands across the ward are chronically exposed, as erratic rainfall and declining soil moisture severely reduce yields of staple crops such as maize, cassava, cowpeas, and green grams, directly threatening food security and household incomes. The mangrove ecosystems located along Takaungu Creek are highly exposed during prolonged dry spells limiting mangrove regeneration affecting fishing activities that sustain some local households. Similarly, the *Kaya* forests and the Kambi ya Shetani forest ecosystem experience drying of understory vegetation, canopy loss, and a heightened risk of human destruction, which accelerates forest degradation and biodiversity loss. Water sources throughout the ward including community pans and shallow wells are highly susceptible to drying up or become saline during prolonged droughts, forcing residents, particularly women and girls, to walk longer distances in search of water, increasing physical strain and reducing time for productive activities. Livestock also face heightened exposure due to diminishing pasture and water availability, resulting in higher mortality rates. This broad and interlinked exposure underscores the close relationship between climate and gendered vulnerability, and livelihood insecurity in the ward, highlighting the need for integrated drought resilience measures that conserve key ecosystems such as Takaungu Creek mangroves and *Kayas* while strengthening community adaptive capacity.

3.4.1.2 Socio-Demographic Vulnerability to Drought

Socio-demographic vulnerability to drought in Mnarani Ward is deeply influenced by changing household dynamics and social inequalities that heighten the exposure of women, youth, and PAD. As droughts become more frequent and prolonged, women have increasingly assumed the role of primary breadwinners, extending their farm work hours to sustain household food supplies amid declining crop yields. With water sources drying up, they are forced to walk long distances in search of water for domestic use and livestock, which increases their physical burden and limits time for income-generating or self-care activities. Youth, on the other hand, face growing social and economic pressures as agricultural livelihoods falter. Many are compelled to engage in casual labor or other short-term income-generating activities to support their families, leading to increased school dropout rates (occurring before they transition into the youth category), early pregnancies and marriages that further entrench intergenerational vulnerability. Persons abled differently experience even greater marginalization during drought periods, often facing neglect and exclusion from community decision-making processes. Their physical and mental incapacities restrict access to food and livelihood opportunities, and minimal participation in adaptive initiatives significantly reduce their resilience. Collectively, these challenges illustrate how drought in the Ward not only undermines livelihoods but also exacerbates social inequalities, underscoring the urgent need for inclusive, gender- and youth-responsive adaptation strategies that address the specific needs of vulnerable groups.

3.4.2 Floods

Flooding is one of the disruptive climate hazards in Mnarani Ward, predominantly associated with intense rainfall episodes and notable periodic events such as El Niño. Historical flood records between 1930 and 2025 indicate major flood occurrences in 1930, 1964–1965, 1992, 1997 and 1998, translating to an average recurrence interval of approximately one significant flood every 10 years (Figure 10). However, flood events have not been as intense after the

2000s, a pattern that can be linked to natural climatic variability. This includes a reduction in the frequency and strength of episodes, shifts in regional rainfall patterns and longer dry intervals between rainy seasons. These changes have moderated extreme precipitation events, thereby reducing the magnitude and spatial extent of flooding across the Ward in recent decades. Vulnerability analysis shows that Midzi Mitsano (53) and Timboni (41) experiences the highest exposure, followed by Vuma (33), Mkwajuni (32) and Vijiweni (27) reflecting both biophysical sensitivity and socio-economic fragility (Figure 11).

The effects of floods on natural and livelihood resources in Mnarani Ward are extensive and interlinked, impacting both terrestrial and marine ecosystems. The Kaya Mnarani forest and surrounding vegetation experience soil erosion, waterlogging, and surface runoff accumulation during heavy rainfall events, which disturb the understory layer and weaken the forest's ecological role in slope stabilization, groundwater recharge, and microclimate regulation. Along the Takaungu Creek, mangrove forests are subjected to prolonged inundation and silt deposition during flooding, leading to smothering of pneumatophores and restricted natural regeneration. This undermines the mangroves' capacity to buffer against coastal erosion and sustain aquatic biodiversity.

Seaweed farming, once a key livelihood activity in Vuma and Takaungu, has since ceased due to both environmental and economic challenges. The formation of a brackish water environment where River Mbogolo meets the Indian Ocean altered salinity levels, rendering the area unsuitable for cultivation. In addition, between 2013 and 2016, strong ocean waves destroyed seaweed farms in Vuma, while in Takaungu, limited knowledge on value addition and the absence of reliable markets led to the collapse of the enterprise. Consequently, recurrent flood events and changing coastal hydrodynamics have further diminished the viability of seaweed farming as a livelihood source within the ward.

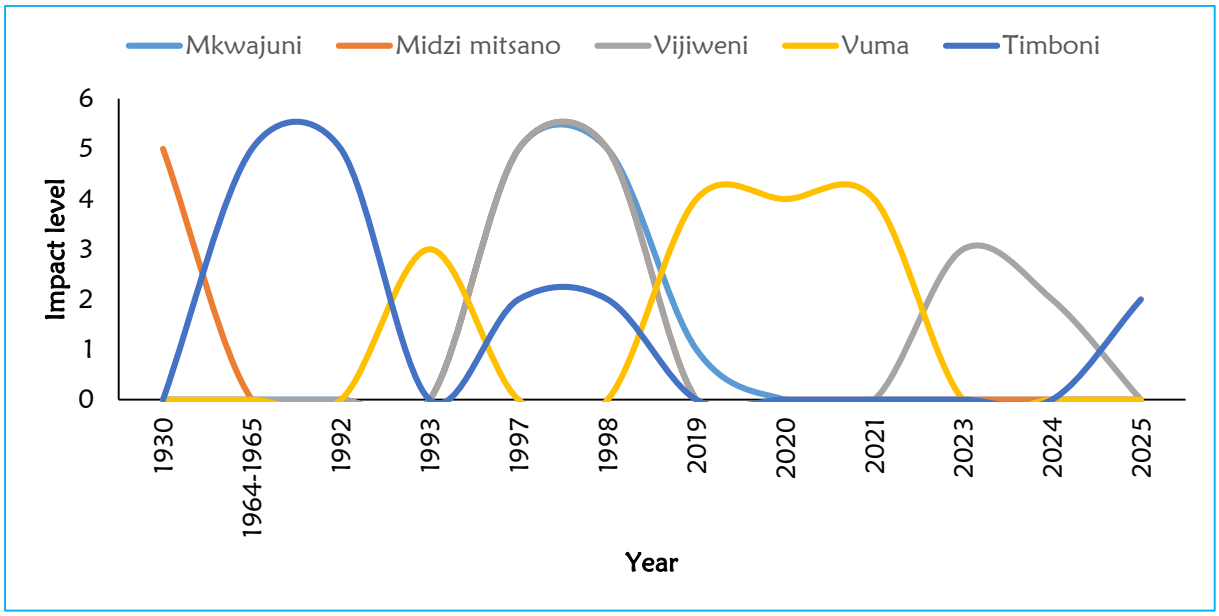

Figure 12: Floods prevalence historical trend in Mnarani (Source: PCVA, 2025)

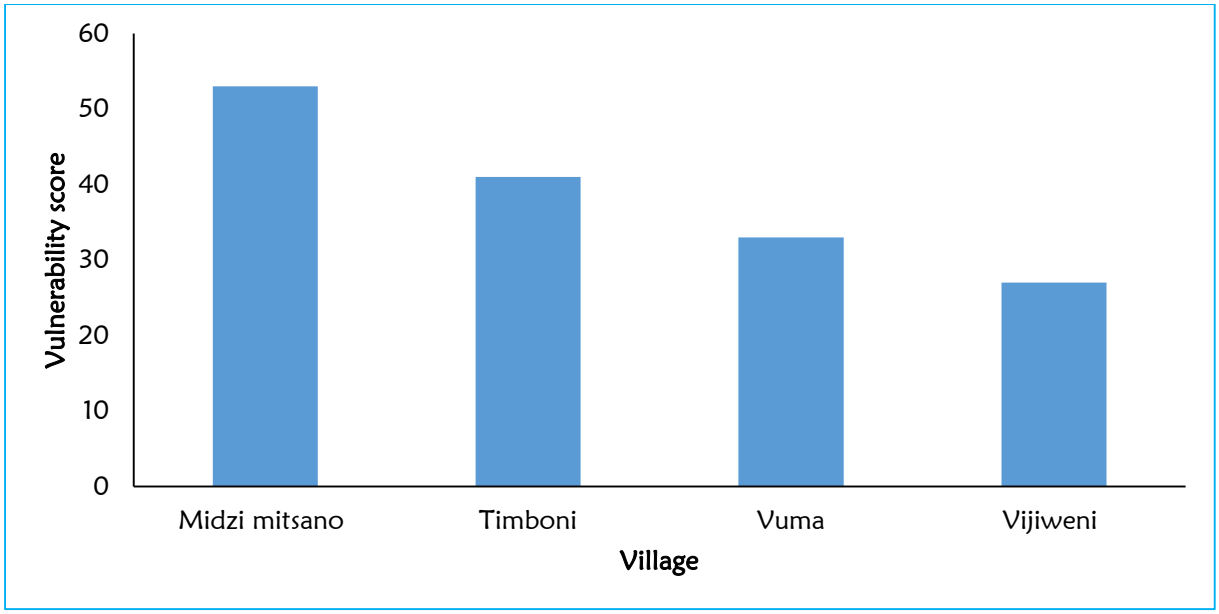

Figure 13: Floods vulnerability score in Mnarani Ward (Source: PCVA,2025)

3.4.2.1 Exposure to floods

Exposure to floods within Mnarani Ward is concentrated in low-lying and hydrologically active zones where settlements and socio-economic activities intersect with natural drainage systems. The most affected areas include Mbogolo, Benesi, Mwakuhenga, Mnarani Town, Majajani, and the coastal-riparian stretches of Takaungu–Kivukoni and Vuma, which host extensive mangrove stands providing critical ecosystem services. However, recurrent flooding and progressive catchment degradation have increased surface runoff, sedimentation, and saltwater intrusion, leading to mangrove die-back, reduced regeneration, and altered species composition. In upland areas such as Midzi Mitsano and Mkwajuni, floodwaters accelerate soil erosion, vegetation loss, and declining terrestrial cover, further reducing groundwater recharge and increasing sediment flow into coastal ecosystems. Poor land-use practices, inadequate drainage infrastructure, and ongoing degradation have collectively determined the frequency and severity of floods, which result in destruction of infrastructure, inundation of farmlands, loss of property and productive assets, and, in severe instances, displacement and loss of life.

3.4.2.2 Socio-Demographic Vulnerability to Pests and Diseases

Socio-demographic vulnerability to floods in Mnarani Ward is influenced by the intersection of social and economic characteristics, including gender roles, age distribution, livelihood systems, and physical ability. Flood impacts are unevenly distributed across social groups, with women, youth, and persons abled differently (PADs) facing the greatest challenges in coping and recovery. Women are particularly affected due to their dual responsibility in income generation and household care. During flood events, they struggle with disrupted market access, contamination of domestic water sources, and the need to safeguard children and household assets. Limited control over land and financial resources further constrains their ability to recover from damages to farms, homes, and small businesses.

Children in flood-prone areas such as Mbogolo, Benesi, Mwakuhenga, Mnarani Town, and Majajani are highly exposed to disease outbreaks, unsafe environments, and learning interruptions when schools are closed or converted into temporary shelters. The elderly and persons abled differently face restricted mobility and heightened dependency during evacuations, often lacking direct access to timely assistance or medical care.

From an economic perspective, households with limited savings or access to credit are often unable to rebuild quickly, perpetuating cycles of poverty and vulnerability. Conversely, a minority of residents with diversified livelihoods, such as formal employment, stable businesses, or external financial support, are better positioned to withstand and recover from flood impacts. Overall, social inequalities, limited economic diversification, and inadequate infrastructure compound vulnerability, leaving marginalized groups in Mnarani disproportionately exposed to the long-term effects of recurrent flooding.

3.4.3 Pests and diseases

Pests and diseases have become a recurrent and increasingly severe climate-related hazard across Mnarani Ward, driven by changes in temperature, humidity, and rainfall patterns. Historical analysis between 1993 and 2024 indicates that outbreaks occur roughly once every three years, cutting across all village (Mkwajuni, Midzi Mitsano, Vijiweni, and Vuma) (Figure 12). The trend demonstrates a gradual intensification over time, with more frequent and widespread occurrences influenced by erratic rainfall, warmer temperatures, and extended wet periods that enhance pest survival and disease transmission. The Participatory Climate Vulnerability Assessment (PCVA) findings indicate that Midzi Mitsano (48) recorded the highest vulnerability score while Vuma (35), Mkwajuni (27), and Vijiweni (12) were affected experiencing recurrent pest infestations, livestock diseases and disruption of livelihoods (Figure 13).

Major outbreaks have been recorded in specific years, notably the 2020 (Corona virus and locust infestation), which caused widespread socio-economic disruption and agricultural losses across the ward. Locust invasions destroyed large areas of cropland, while the COVID-19 pandemic severely constrained household incomes and access to farm inputs. Moreover. a new wave of health challenges emerged with an outbreak of red eye, alongside increased incidences of cholera, chikungunya, and respiratory infections, particularly during humid post-flood or dry conditions. Agricultural assessments highlight persistent infestations of whiteflies, armyworms, and fungal leaf blight, which have reduced yields of maize. Simultaneously, livestock diseases, including Foot and Mouth Disease (FMD), Newcastle disease, and East Coast fever, have pause a challenge, undermining food security and livestock productivity. These challenges have been aggravated by limited access to veterinary services and inadequate pest management practices.

The Kambi ya Shetani Kaya and mangrove areas, have also shown increasing vulnerability to climate-related pest and disease pressures. The Kaya forests, which provide vital ecosystem services such as soil stabilization, medicinal plants, and microclimate regulation, have experienced reduced regeneration capacity due plant diseases and pests. Along the coastline, mangrove ecosystems have been affected by fungal rot, particularly during prolonged humid and warm periods. These disturbances not only threaten biodiversity and coastal protection but also disrupt fish breeding and nursery habitats, undermining the ecological and economic balance that sustains local communities.

At the community level, exposure to pests and diseases is strongly shaped by socio-economic status and gender roles. Women and youth, dependent in agricultural and livestock practices face disproportionate impacts from declining yields and recurrent health burdens. Women, in particular, experience increased workloads during outbreak periods, engaging in pest control, replanting and caregiving while their income opportunities shrink. Collectively, these challenges highlight how pest and disease outbreaks, compounded by climate variability, continue to undermine livelihoods, ecosystem stability, and gender equity across Mnarani Ward.

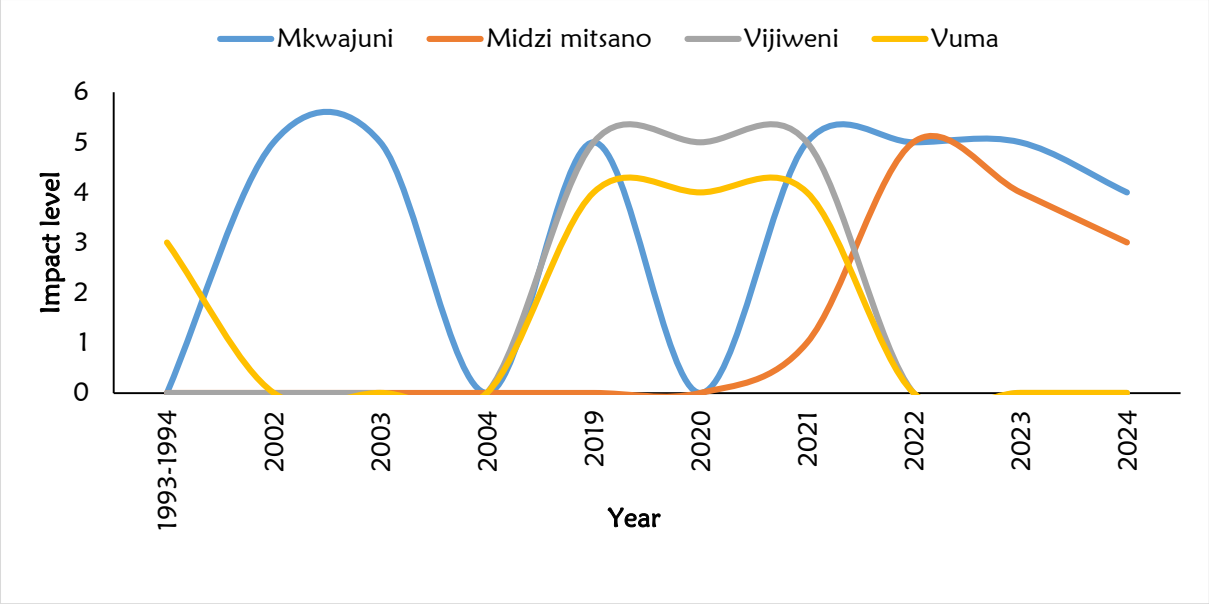

Figure 14: Pests and diseases prevalence historical trend in Mnarani (Source: PCVA, 2025)

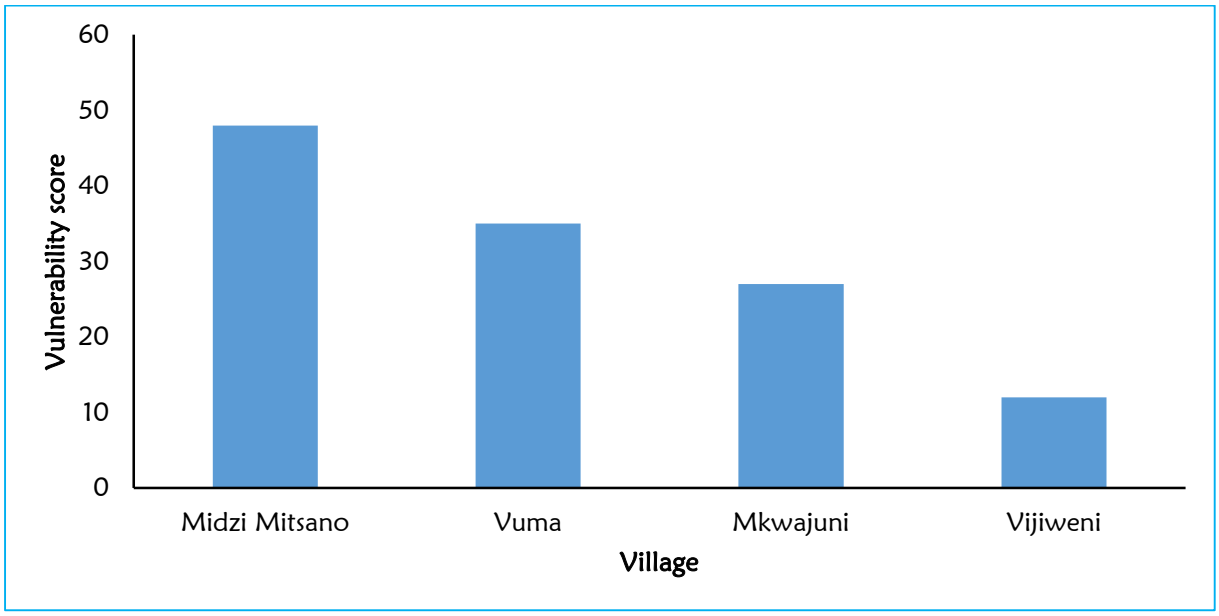

Figure 15: Pests and diseases vulnerability score in Mnarani Ward (Source: PCVA,2025)

3.4.3.1 Exposure to pest and diseases

Exposure to pests and diseases in Mnarani Ward is widespread and cuts across the ward. The entire ward is vulnerable due to its dependence on climate-sensitive livelihoods such as crop production, livestock keeping, and small-scale fisheries. Shifting climatic conditions, characterized by erratic rainfall, prolonged wet spells, and rising temperatures, create favorable environments for pest outbreaks and the spread of diseases. Mkwajuni, Midzi Mitsano, Vijaweni, and Vuma all experience recurring infestations of crop pests such as whiteflies, armyworms, and fungal blight, while livestock diseases including Foot and Mouth, Newcastle, and East Coast fever remain prevalent throughout the ward. Increased mosquito populations during wet seasons have also heightened exposure to malaria, chikungunya, and other vector-borne diseases. The combination of climatic stress, ecological degradation, and reliance on natural resources makes exposure to pests and diseases a ward-wide challenge that undermines both livelihoods and health security.

3.4.3.2 Socio-Demographic Vulnerability to Pests and Diseases

Pests and diseases present significant socio-demographic vulnerabilities across all population groups within the Ward, amplifying existing inequalities and eroding household resilience. Common diseases such as malaria, chikungunya, red-eye infections, and cholera have widespread health and economic consequences. Women are particularly burdened due to their central caregiving roles, often tending to sick children and family members while maintaining domestic and livelihood responsibilities. This dual strain limits their productivity and income, generating potential, deepening household poverty. The situation is especially critical in female-headed households, where women shoulder both caregiving and sole economic responsibilities, leaving them with minimal capacity to respond to or recover from disease outbreaks.

Pest infestations, particularly armyworm, have devastated farmlands, reducing crop yields and threatening food security for agriculture-dependent households. Diminished harvests have led to increased food prices, nutritional deficits, and heightened vulnerability among low-income families. Livestock diseases such as foot-and-mouth disease and constipation have disrupted pastoral livelihoods, forcing men to trek long distances in search of pasture and water. Youth are often drawn into these efforts or resort to alternative, sometimes unsustainable, livelihood activities as agricultural opportunities decline.

Elderly persons face increased susceptibility to both pests and diseases due to reduced immunity and limited access to healthcare. Persons Abled Differently (PADs) are also at high risk, as mobility challenges and inadequate inclusion in adaptive programs restrict their ability to engage in pest management activities or relocate during severe infestations and disease outbreaks. The combined effects of recurring pest invasions, livestock diseases, and vector-borne illnesses have weakened community resilience, depleted household resources, and heightened vulnerability, particularly among female-headed and low-income households.

3.4.4 Environmental degradation

Environmental degradation has persisted as a major climate-related concern in Takaungu, Vuma and Timboni areas, driven by increasing demand for fuelwood, fisheries products, tourism development, urban expansion, and other resource-based activities. These pressures have intensified deforestation within both mangrove and terrestrial forest ecosystems, accelerating the loss of critical habitats and ecological integrity. Historical assessments reveal a consistent pattern of ecosystem decline, with the period between 2005 and 2023 recording the most severe degradation episodes (Figure 14). More recently, rapid population growth, poor waste management practices, and unregulated dumping have contributed to escalating pollution levels, debris accumulation, and the deterioration of aquatic and coastal ecosystems. Vulnerability assessments indicate that Takaungu (72), Timboni (43) and Vijiweni (19) were among the most affected localities (Figure 15).

The impacts of environmental degradation are extensive and multidimensional. Notable effects include the loss of mangrove cover, siltation of water bodies, declining fish stocks, and erosion of traditional livelihoods, all contributing to reduced household incomes and heightened vulnerability. Additionally, the destruction of fish breeding and feeding grounds, coupled with intensified coastal and beach erosion, has diminished the area's capacity for carbon sequestration and natural shoreline protection. The cumulative loss of ecosystem services has weakened community resilience and constrained opportunities for sustainable income generation, posing significant challenges to both socio-economic development and environmental sustainability

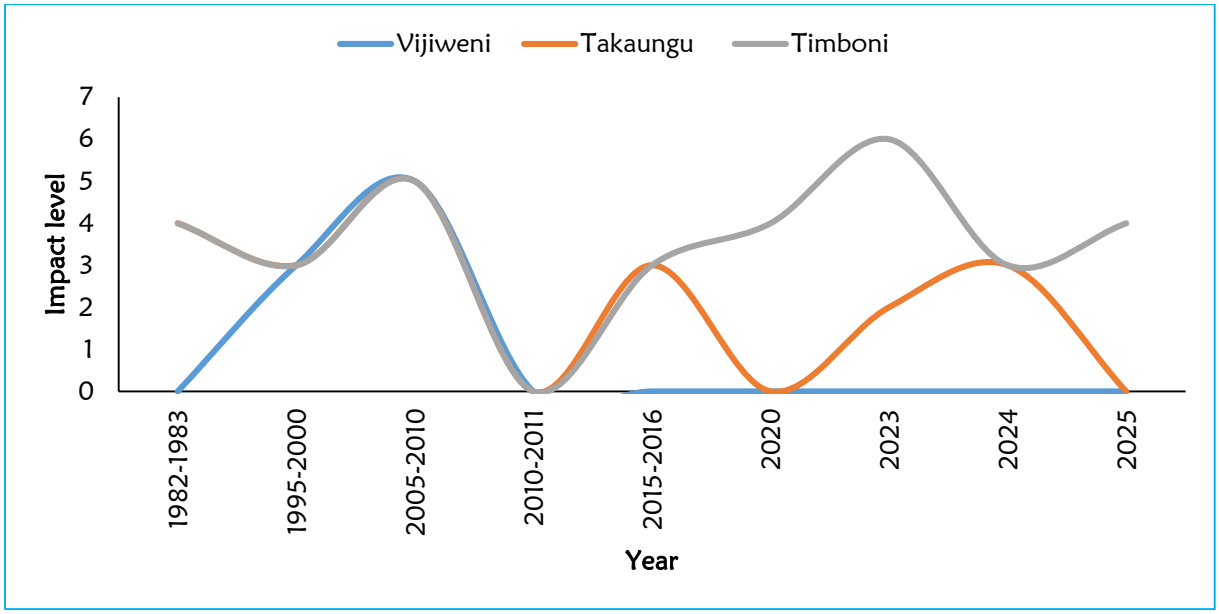

Figure 16: Environmental Degradation prevalence historical trend in Mnarani (Source: PCVA,2025)

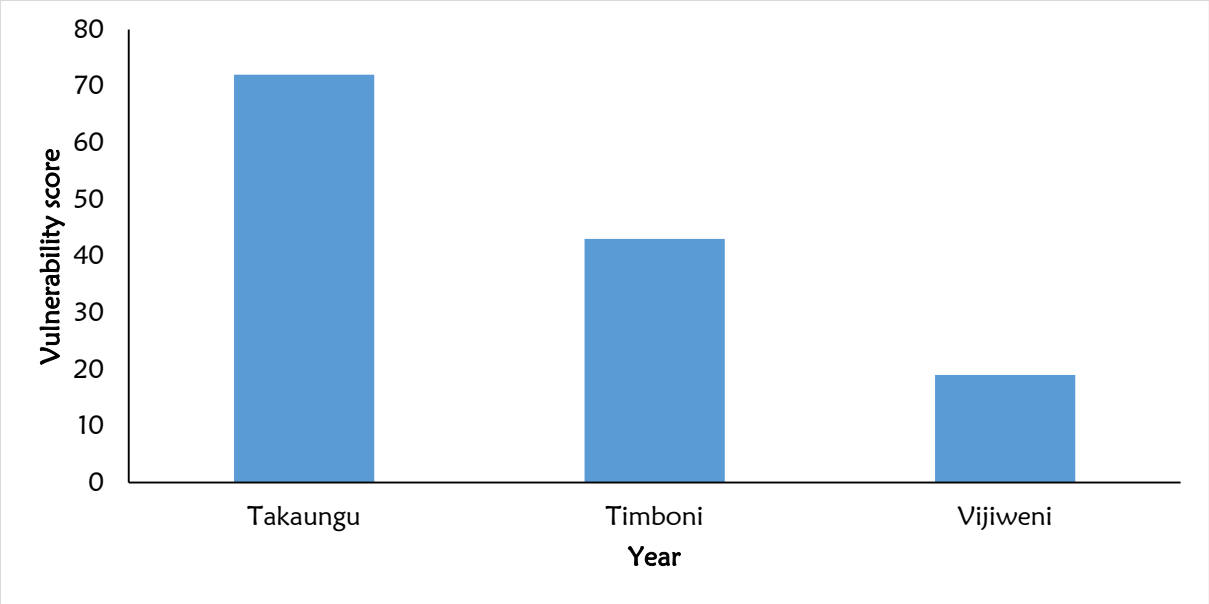

Figure 17: Environmental degradation vulnerability score in Mnarani Ward (Source: PCVA,2025)

3.4.4.1 Exposure to Environmental degradation

Exposure to environmental degradation in Takaungu, Vuma, and Timboni is strongly linked to the loss and disturbance of mangrove ecosystems. Households living adjacent to mangrove area are directly exposed to the impacts of deforestation, sedimentation, and pollution that reduce the ecological health of these habitats. Uncontrolled cutting of mangroves for fuelwood and construction materials, combined with the expansion of settlements and fish landing sites into sensitive areas, has increased human ecosystem interaction and heightened degradation pressure. The decline of mangrove cover has also reduced the natural buffering capacity against tidal surges and coastal erosion, exposing households and fishing infrastructure to greater physical and economic risks. In areas, where mangrove-dependent activities such as fishing and crab harvesting are central to livelihoods, the loss of mangrove resources directly threatens household income and food security.

3.4.4.2 Socio-Demographic Vulnerability to Environmental Degradation

Socio-demographic vulnerability to mangrove degradation is particularly high among communities whose livelihoods depend on coastal and nearshore resources. Women, who rely on fish processing, experience reduced earnings as fisheries stocks decline thus affecting their livelihoods. Youth and poorer households, lacking alternative income options, are more likely to engage in unsustainable harvesting practices, perpetuating degradation and dependency. In areas such as Timboni and Vijiweni, limited environmental awareness and inadequate community-based management have further constrained adaptive capacity. Overall, the degradation of mangrove ecosystems has weakened both environmental resilience and community livelihoods, deepening social and economic vulnerability across the coastal zone.

3.4.5 Strong winds

Strong winds have emerged as a significant climatic hazard, with Vuma being the most affected area with the earliest event in 1992 (Figure 16). The phenomenon has been exacerbated by extensive mining activities undertaken by Mombasa Cement and widespread deforestation, which have left vast areas of land bare and highly susceptible to wind erosion and turbulence with peak levels recorded 2016 to 2017. The absence of vegetative cover has increased the exposure of settlements and infrastructure to strong wind impacts.

In 2023, the area experienced severe whirlwind events, resulting in extensive damage to infrastructure and productive systems. Critical facilities such as schools, farmlands, and coconut plantations were among the most severely affected, while settlements and road networks also sustained moderate damage. The socio-economic implications are particularly pronounced for households, who disproportionately bear the burden of recovery and adaptation. They are often compelled to engage in repair work, manage household economic losses, and navigate increased exposure to hazards arising from damaged structures and disrupted livelihoods. This heightened vulnerability underscores the gendered dimensions of climate-induced wind hazards and the urgent need for integrated mitigation and adaptive interventions in Vuma and surrounding areas.

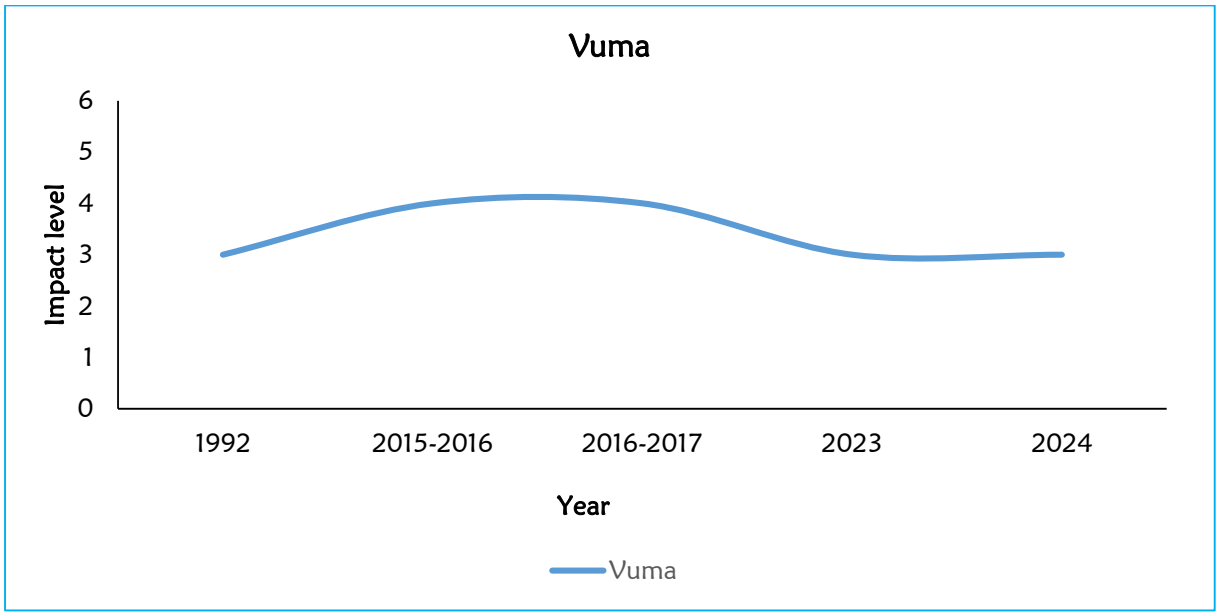

Figure 18: Strong winds historical prevalence historical trend in Mnarani (Source: PCVA,2025)

3.4.5.1 Exposure to Strong Winds

Exposure to strong winds in Vuma is concentrated mainly around open mining zones and upland coastal ridges, where deforestation and excavation have removed natural vegetation

buffers that once reduced wind velocity. These windstorms, occurring approximately once every six years, have intensified due to land degradation and the loss of terrestrial cover. The effects are severe on both ecosystems and community assets eroding soils as well as damaging farmlands and plantations, particularly coconut and cashew crops. Settlements and public facilities such as schools located in exposed areas frequently experience roof damage and structural weakening during high-wind events. Women and youth engaged in small-scale farming and informal trade are among the most exposed, as strong winds destroy drying racks, market stalls, and storage structures, resulting in direct livelihood disruptions and economic losses.

3.4.5.2 Socio-Demographic Vulnerability to Strong Winds

Socio-demographic vulnerability to strong winds in Vuma varies across gender, livelihood type, and economic capacity. Women and youth, who are primarily engaged in small-scale coconut processing, farming, and informal trade, are disproportionately affected as strong winds destroy market stalls, drying racks, and storage sheds. Recovery is further constrained by limited access to credit and savings, increasing household dependency on unstable income sources. Women-headed households bear additional burdens of repair and caregiving during displacement periods, reflecting the gendered nature of post-disaster recovery.

Men, particularly those working in artisanal mining and fishing, face physical risks during wind events due to collapsing debris and turbulent sea conditions. Children and elderly persons are indirectly affected through damage to schools and homes, disrupting education and access to essential services. Persons abled differently (PADs) and low-income households are also at greater risk, lacking the resources or mobility needed to reinforce their homes or relocate temporarily during high wind periods.

Recurrent wind-related destruction has gradually weakened household resilience and strained community coping systems. The cycle of damage and repair has led to declining productivity and economic insecurity, particularly among smallholder farmers and low-income earners. To reduce vulnerability, integrated interventions are required such as vegetation restoration, enforcement of mine rehabilitation measures and promotion of wind-resistant construction. These measures would strengthen both ecosystem stability and community resilience against intensifying wind hazards in Vuma and the broader Mnarani Ward.

3.4.6 Coral bleaching

Coral bleaching events in Timboni have increased noticeably in both frequency and severity over the years. During the late 1990s and early 2000s, bleaching incidents were sporadic and localized, primarily linked to episodic rises in sea surface temperature (Figure 17). However, since 2020, bleaching has become more frequent and widespread, occurring across multiple reef sites and lasting for extended periods. By 2025, observations indicate that bleaching is now a recurrent environmental hazard, reflecting the intensifying impacts of climate-induced ocean warming and reduced ecosystem resilience. The trend underscores a transition from isolated occurrences to a persistent stress regime affecting coral health and reef functionality.

The increasing frequency of bleaching has had profound ecological consequences for Timboni's coral reef ecosystem. Repeated exposure to thermal stress has led to widespread coral mortality, loss of live coral cover, and a gradual shift in reef community composition. The decline in coral diversity has diminished habitat complexity, affecting reef-associated fish and invertebrate populations. Sediment runoff, nutrient enrichment, and pollution from land-based activities have further weakened coral recovery, while algal overgrowth has become more pronounced in degraded zones. As a result, the structural integrity and ecological productivity of the reef

have been significantly reduced, compromising its role as a breeding ground, carbon sink, and natural barrier against coastal erosion.
Coral bleaching has equally critical implications for local livelihoods and coastal safety. The deterioration of reef ecosystems has reduced fisheries productivity, directly affecting small-scale fishers who rely on reef species for income and food security. The decline in coral health has also eroded the area's tourism potential, limiting opportunities for snorkeling, diving, and eco-based enterprises that previously supported community livelihoods. Furthermore, the loss of coral structures has weakened natural shoreline protection, making coastal settlements more vulnerable to wave action, storm surges, and erosion. Men, who are heavily involved in fisheries activities, face increased economic strain and reduced adaptive capacity as a result of these cumulative impacts.

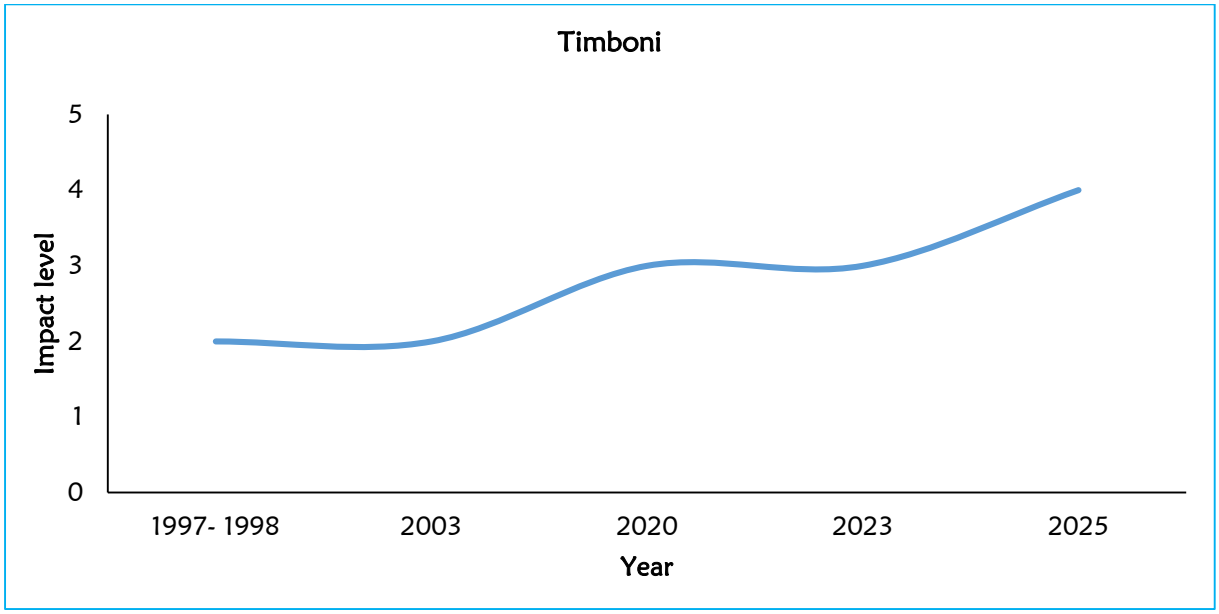

Figure 19: Coral bleaching historical prevalence historical trend in Mnarani (Source: PCVA,2025)

3.4.6.1 Exposure to coral bleaching

Recurrent bleaching episodes have weakened reef structure, reduced live coral cover, and impaired ecological functionality. The degradation has also disrupted key ecosystem services such as fish habitat provision, shoreline protection, and nutrient cycling. The cumulative effect of these stressors signals a shift from isolated thermal anomalies to a persistent bleaching hazard driven by climate change, water pollution, and unsustainable coastal activities. Overall, exposure to coral bleaching in Timboni is now widespread and poses a serious threat to both ecological stability and community livelihoods

3.4.6.2 Socio-Demographic Vulnerability to coral bleaching

The vulnerability of Timboni's population to coral bleaching is strongly linked to livelihood dependence, gender roles, and economic capacity. A significant proportion of men in the community engage in small-scale fishing, while women are actively involved in fish processing, trading, and related activities that depend on healthy coral ecosystems. As coral degradation intensifies, these groups face declining fish stocks, reduced catches, and lower household incomes, which heighten economic insecurity and food vulnerability. Limited livelihood diversification and minimal access to credit or modern fishing technologies further constrain the community's adaptive capacity. The loss of natural reef barriers has exposed coastal settlements and public infrastructure to erosion and wave impacts. In addition, reduced marine productivity has disrupted traditional fishing patterns, intensifying competition over dwindling resources. Overall, coral bleaching in Timboni has deepened socio-economic vulnerability, affecting men,

and women differently but collectively undermining the resilience of households that rely heavily on marine and coastal resources

3.5 Classification of Exposure and Resilience through Livelihoods

The ward is characterised by livelihoods that are highly exposed to environmental, economic and institutional shocks, while its capacity for resilience remains constrained by limited income diversity, low margins and persistent poverty. The dominant cash-income sources are informal and closely tied to marine/coastal and transport services with modest supplementary incomes from small-scale business and agriculture, fishing, wine tapping and livestock. At the same time, county data indicate that Kilifi poverty headcount has been around 46.1 per cent, higher than the national average. These factors combine to yield a system with narrow buffers and high vulnerability, thereby demanding careful classification of exposure and resilience through the livelihood lens.

3.5.1 Livelihood Structure and Monthly Incomes

Table 2 presents the estimated monthly income levels (KES) across six villages; Takaungu, Vuma, Vijiweni, Mikwajuni, Timboni, and Midzi Mitsano, demonstrating the distribution of economic activities within the ward. Wine tapping represents the highest-income activity, with monthly earnings ranging from KES 36,000 to KES 120,000. Fishing and mining each generate approximately KES 50,000 per month, while livestock production yields a moderate and relatively stable income between KES 30,000 and KES 50,000. Collectively, these activities constitute the principal sources of household income and are central to the local economic structure Coral block harvesting, undertaken predominantly by private entities, produced an estimated KES 2,770,000 in total output monthly.

The high income derived from wine tapping is primarily attributed to strong and consistent market demand, particularly in coastal regions where palm wine is both a commercial and cultural commodity. The activity requires minimal input costs once the trees mature, allowing for sustained profitability with limited recurrent expenses. Additionally, wine tapping provides a continuous year-round income stream, unlike other seasonal livelihoods such as crop farming or fishing. The availability of mature coconut and palm trees, coupled with available ready market, further enhances its profitability. Collectively, these factors contribute to wine tapping's position as one of the most lucrative livelihood activities in the ward.

Mining serves as an important source of income for local communities; however, it remains highly susceptible to climate-induced hazards such as flooding as well contributing to severity of some hazard's i.e strong winds. These events, combined with unregulated and excessive extraction, accelerate land degradation, erosion, water contamination, and the loss of terrestrial habitats. The situation underscores the necessity for adopting sustainable and regulated mining practices aimed at minimizing environmental damage and safeguarding the livelihoods that rely on these natural systems.

Conversely, livelihood activities such as charcoal and firewood trading, small-scale vending, poultry rearing, nursery production, and subsistence farming generate comparatively low monthly incomes, typically ranging from KES 2,500 to KES 8,000. These enterprises are largely dominated by women and youth and are constrained by limited capital, narrow market access, and pronounced vulnerability to seasonal fluctuations. As a result, participants often depend on informal financial mechanisms such as chamas, SACCOs, and microcredit schemes to sustain operations. Households reliant on these low-return activities generally exhibit weak financial resilience, with limited capacity to save, reinvest, or withstand economic shocks.

Table 2: Monthly earnings by livelihood and village (Source: PCVA, 2025)

Livelihood	Income (KES)					
	Mkwajuni	Vijiweni	Midzi Mitsano	Vuma	Timboni	Takaungu
Agriculture	9,000	8,000	10,000	20,000		
Fishing	50,000	11,500		50,000	10,000	
Small scale business	15,000	20,000	15,000	20,000	15,000	20,000
Livestock production	30,000	19,000	50,000	42,000	1,200	15,000
Wine tapping	60,000	60,000	126,000	78,000	36,000	
Mining		50,000				
Tree nurseries/Seedlings		9,000	7,500			4,000
Remittance	10,000			20,000	15,000	10,000
Chamas/Credit services/SACCO	3,000				10,000	6,000
Coral blocks (Owner)				2,770,000		
Wages	30,000		30,000		12,000	18,000
Transport	20,000		42,000			
Stipend					12,000	4,000
Charcoal/Firewood					2,500	
Transport (Motorbike)						12,000

The poverty profile of the reflects the economic fragility of its livelihoods. Field data show that many households in all six villages experienced varied income levels to meet their basic needs over the past year. Figure 18 below presents the number of people who reported lacking money to meet critical needs such as food, housing, clothing, transport, and medication in the last 12 months.

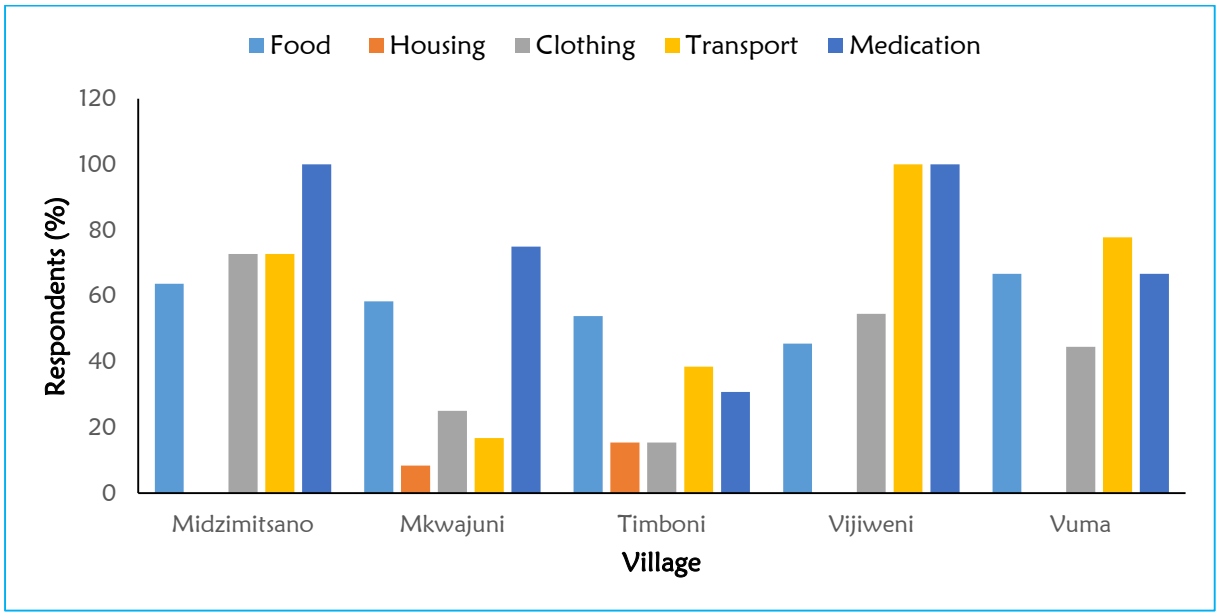

Figure 20: Respondents who lacked money to Meet Basic Needs in the Last 12 Months (Source: PCVA 2025)

This data provides evidence of widespread deprivation across the ward. The data reveals disparities in household access to essential needs across the six villages. Access levels were highest for medication and food, yet still insufficient to meet community needs comprehensively. Medication access ranged between 30.8% in Timboni and 100% in Midzimitsano and Vijijweni, suggesting variable health service availability and affordability. Food access averaged between 45.5% and 66.7%, highlighting persistent livelihood pressures linked to low income and environmental stressors. Transport and clothing access also reflected strong inter-village disparities, with higher values in Midzimitsano (72.7%) and Vuma (77.8%) depicting broader structural poverty.

This economic marginalization reinforces a persistent cycle of low resilience, as households with limited financial resources are unable to invest in improving livelihoods, replacing lost assets, or adopting climate adaptation measures. As a result, every climatic, health, or economic shock exacerbates their vulnerability, often compelling families to resort to adverse coping mechanisms such as cutting down meal portions, selling productive assets, or withdrawing children from school to engage in labour.

3.5.2 Vulnerabilities of Livelihood Options

All livelihood options in the ward while essential for household sustenance, expose the community to varying degrees of risk. Wine tapping, though offering the highest cash income, are highly sensitive to fluctuations climatic conditions where during rainy season such activities get to be affected. Heavy rains and flooding disrupt local alcohol brewing by hindering fermentation, and limiting access to raw materials and markets. Conversely, extreme drought conditions reduce coconut sap availability as trees dry up, sharply lowering production. Together, these climatic extremes create recurring periods of low productivity and income loss, making brewing a highly climate-sensitive.

Fishing, though a vital source of income and food security in the ward, faces increasing exposure to both environmental and institutional challenges. Overexploitation of marine resources, destruction of mangrove habitats, and the widespread use of unsustainable fishing gear have led to a steady decline in fish availability. Shifts in ocean currents and the growing frequency of rough sea conditions further restrict fishing days and reduce overall catch. These ecological constraints are intensified by inadequate supporting amenities, which cause post-harvest losses, and by weak governance within Beach Management Units (BMUs), limiting enforcement of sustainable practices. As a result, the fishing sector is experiencing diminishing returns, heightening livelihood vulnerability and threatening long-term resource sustainability

Micro-enterprise and trade activities, practiced mainly by women and youth, offer minimal financial returns and are constrained by limited capital, seasonal demand, and competition. Drought deeply affects both micro-enterprise and agriculture by reducing household incomes, shrinking local demand, and disrupting market activity. In farming, it causes crop failure, poor yields, and loss of raw materials essential for small-scale trade. Water scarcity and declining livestock productivity further strain livelihoods, making drought a major driver of food insecurity and economic instability in the ward.

Auxiliary activities such as mining, transport, livestock keeping, coral block making, and tree-seedling production provide crucial livelihood diversification for households; however, these ventures remain highly fragile due to environmental, economic, and technical constraints. Mining and coral block making are limited by resource depletion and regulatory restrictions, while transport services are vulnerable to fluctuating fuel prices and poor road infrastructure. Livestock keeping faces recurrent disease outbreaks and water scarcity, whereas tree-seedling

production is highly sensitive to drought and requires sustained care and inputs. Overall, these activities offer short-term income opportunities but lack resilience to climatic and market shocks.

Underlying all these vulnerabilities is the persistent issue of poverty. The low income levels indicate that most households operate within or just above subsistence thresholds, leaving little room for investment or shock recovery. In this context, even minor external disturbances, such as a week of poor fishing weather, a spike in fuel costs, or an illness in the family can severely undermine household stability. The reliance on informal, low-profit activities without access to formal financial services amplifies this fragility.

3.5.3 Overall Classification and Implications

The cumulative evidence positions Mnarani Ward within a high-exposure, low-to-moderate-resilience category. The community's economic dependence on climate-sensitive and informal livelihoods, combined with high poverty levels and gendered unpaid work burdens, restricts adaptive capacity and heightens vulnerability. Addressing these constraints requires integrated measures that improve income stability in high-risk sectors, promote diversification into higher-value chains, reduce women's unpaid workload through labour-saving interventions, and strengthen social safety nets. Expanding access to microfinance, improving infrastructure such as roads and cold storage, and investing in ecosystem-based adaptation particularly mangrove restoration and sustainable fisheries will be vital in shifting the ward's resilience profile from reactive coping to proactive adaptation.

3.6 Analyzing causes of vulnerability.

3.6.1 Classification of Severity of the Vulnerabilities and Their Causes

The ward exhibits varying degrees of vulnerability across its six sampled villages; Takaungu, Vuma, Vijiweni, Mikwajuni, Timboni and Midzi Mitsano, influenced by ecological setting, livelihood dependence, and adaptive capacity. The classification was informed by participatory scoring and spatial analysis. Mikwanjuni, Vijiweni, Timboni, Takaungu, and Vuma demonstrated high vulnerability due to their proximity to the coastline, dependence on marine resources and exposure to saline intrusion and floods. Midzi Mitsano was classified as moderately vulnerable, where drought, floods, pest infestation, and limited water access threaten agricultural productivity.

Vulnerability is compounded by limited livelihood diversification, chronic poverty, and gendered unpaid labour, which reduces women's capacity to respond to shocks. Resource-dependent livelihoods in the ward are highly exposed to climatic and ecological stresses, while socio-economic constraints limit adaptive capacity. This classification highlights the multidimensional nature of vulnerability, where ecological, economic, and gender factors intersect to determine overall risk.

3.6.2 Identification of Unsafe Conditions

Unsafe conditions in the ward refer to the immediate settings and circumstances that heighten exposure to climate-related hazards. Settlements located along tidal zones in Mikwanjuni, Vijiweni, Timboni, Takaungu, and Vumba experience recurrent flooding and saltwater intrusion, resulting in frequent loss of assets, reduced agricultural productivity, and disruption of fishing and small-scale trade. Inadequate drainage systems and poor solid waste management further exacerbate flood impacts, while deforested areas around the mangrove belt expose the coastline to erosion, storm surges, and loss of protective natural buffers.

From a gender perspective, unsafe conditions are reflected in the disproportionate burden placed on women and girls, who must travel longer distances to collect water and firewood during dry seasons often through degraded and unsafe landscapes. Limited access to climate-

resilient infrastructure, including raised storage facilities, safe fish landing sites, and wind-resistant shelters, intensifies vulnerability for both men and women, but particularly affects households headed by widows, youth, and Persons Abled Differently (PADs).

Additionally, the absence of proper land-use planning and weak enforcement of building codes have encouraged informal settlements in flood-prone areas, while limited disaster preparedness mechanisms and early warning systems leave communities ill-equipped to respond effectively. These factors collectively compound vulnerability by linking environmental degradation, poverty, and inadequate institutional capacity.

3.6.3 Identification of Dynamic Pressures or Determinants of Vulnerability

Dynamic pressures represent the intermediate processes through which underlying socio-economic and institutional weaknesses are translated into direct, unsafe conditions that heighten vulnerability. In Mnarani Ward, these pressures manifest through rapid population growth, unregulated land-use change, overexploitation of natural resources, and weak enforcement of environmental by-laws, all of which interact to degrade ecosystems and undermine resilience.

A clear manifestation of these dynamic pressures is observed in the ongoing mangrove degradation in Takaungu. The degradation is extensive across the entire forest area, with the most significant losses recorded in Takaungu Sub-location. Local accounts indicate that large-scale clearing of mangroves occurred in the Kiriba/Wangwani Sub-location, where the Kenya National Highways Authority (KeNHA) cleared mangroves to facilitate road construction. This activity not only caused the direct loss of mangrove cover but also led to increased sedimentation in adjacent mangrove zones within Takaungu Sub-location. The resulting sediment deposition has further altered the habitat structure, accelerating the decline of mangrove health and coverage. Consequently, the reduction in mangrove cover has contributed to declining fish stocks, compelling local households to diversify into alternative yet resource-dependent coping strategies, including the extraction and sale of mangrove poles. These adaptive but unsustainable responses perpetuate ecological degradation, creating a feedback loop between livelihood stress and environmental decline.

From an institutional perspective, weak coordination between county departments, local community structures, and national agencies constrains effective resource governance. Enforcement of conservation by-laws remains sporadic, while early warning systems for floods and droughts are either absent or poorly operationalized, reducing preparedness and timely response. Economically, limited access to credit facilities continue to undermine household income stability. On the social front, the exclusion of omen from decision-making forums, particularly in fisheries management committees, limits the inclusivity and sustainability of adaptation initiatives.

3.6.4 Identification of Underlying Causes of Vulnerability

The underlying causes of vulnerability in Mnarani Ward stem from interlinked socio-economic, environmental, and institutional deficiencies that collectively erode household resilience and adaptive capacity. Structural poverty remains pervasive, limiting access to education, technology, financial services, and diversified livelihood opportunities. According to the Participatory Climate Vulnerability Assessment (PCVA, 2025), the average household size in Mnarani Ward is 7 persons per household, indicating a high dependency ratio and increased pressure on limited natural and financial resources. Larger households require greater quantities of food, water, and energy, yet often rely on subsistence livelihoods and ecosystem-dependent activities such as fishing and small-scale agriculture. This dependency entrenches poverty cycles and increases exposure to environmental and economic shocks.

Environmental degradation constitutes another major underlying factor, manifesting in mangrove deforestation, soil erosion and declining marine productivity. The loss of vegetative cover reduces the capacity of ecosystems to buffer against climate-induced hazards, particularly floods, droughts, and strong winds, thereby increasing community exposure and reducing long-term ecological stability.

Institutionally, weak governance frameworks and limited coordination between the County Government of Kilifi, the Ward Climate Change Planning Committee (WCCPC), and relevant national agencies hinder the implementation of adaptive interventions. Resource constraints, overlapping mandates, and inadequate policy enforcement reduce the effectiveness of climate action at both ward and county levels. Furthermore, the absence of comprehensive early warning and information dissemination systems results in delayed community responses to climatic hazards, shifting the focus from preparedness to crisis management.

Gender inequality further compounds vulnerability. Patriarchal norms and customary land tenure systems restrict women's ownership and control of productive assets, constraining their participation in decision-making and their capacity to adopt climate-smart technologies or access credit facilities. Consequently, women remain overrepresented in low-return and climate-sensitive livelihood activities, limiting their economic independence and adaptive potential.

3.6.5 Prioritization of Causes

Prioritization of the identified causes was achieved through participatory ranking during the PCVA exercises, considering frequency, severity, and reversibility of impacts. The highest priority causes were poverty, environmental degradation, and weak institutional support systems. Poverty restricts adaptive investments and perpetuates reliance on climate-sensitive livelihoods, while environmental degradation reduces the ecosystem's regenerative capacity. Weak institutional mechanisms delay response and hinder the implementation of ward-level adaptation initiatives.

Medium-priority causes include gender inequality, population pressure, and lack of access to climate information, all of which significantly influence vulnerability dynamics. Addressing these priority causes requires an integrated approach, strengthening governance structures, enhancing livelihood diversification, and promoting gender-responsive adaptation strategies. Interventions such as mangrove restoration, provision of microcredit for women's enterprises, and establishment of community-based early warning systems are essential to reduce long-term vulnerability and build resilience across all villages in the ward.

4 Chapter 4: Gender/Age Vulnerability Analysis for Climate Hazards

Gender and age critically shape exposure, sensitivity, and adaptive capacity across population groups in Mnarani Ward, where climate hazards, including drought, floods, pests and diseases, environmental degradation from mining by Mombasa Cement, strong winds intensified by deforestation for mining, and coral bleaching, intersect with entrenched socio-cultural inequalities and poverty dynamics (Table 1). These climate threats disproportionately affect women, girls, youth, elderly persons, and Persons Abled Differently (PADs), whose vulnerability is reinforced by unequal access to resources, gendered labour divisions, mobility restrictions, and limited participation in household and community-level decision-making (UN Women, 2022; IPCC, 2022).

A key vulnerability factor among women and girls in Mnarani is the heavy, undervalued burden of unpaid domestic and care work, including childcare, cooking, water collection, firewood gathering, laundry, elder care, cleaning, and subsistence farming. During PCVA assessments, this workload was valued at KES 1,000 per day, yet remains largely unrecognised in formal economic structures. Nationally, women contribute over 70% of unpaid care labour (KNBS, 2023). Climate hazards worsen this workload by extending the time required to secure water, food, fuelwood, and healthcare, deepening "time poverty" and limiting women's ability to engage in paid work, capacity-building, leadership, and climate adaptation activities (ILO, 2018; UN Women, 2023).

Within Mnarani Ward, gendered power imbalances further exacerbate vulnerability. Many women lack decision-making authority over land, farming inputs, fishing investments, asset use, or household finances. This lack of agency limits their ability to adopt climate-resilient technologies or participate effectively in adaptation planning. Climate-related stress often heightens intra-household tensions, contributing to gender-based violence (GBV), including emotional, physical, economic, and sexual abuse, which reduces women's mobility, undermines economic independence, and inhibits participation in public life. The recent rise in femicide cases in Kenya has deepened fear and restricted women's movement, further weakening adaptive capacity (UNFPA, 2023; Kenya Police Service, 2024).

Drought is among the most severe hazards in Mnarani Ward, intensifying water scarcity and reducing agricultural productivity. Women and girls walk longer distances, often up to 5 kilometres, to access water and firewood, exposing them to fatigue, heat stress, dehydration, musculoskeletal injuries, and protection risks such as harassment and assault. Female-headed households are particularly vulnerable due to limited labour support and financial constraints. Men face livelihood losses from poor crop yields, declining forest resources, and reduced fishing opportunities, resulting in psychological stress. Youth experience reduced opportunities for casual labour and apprenticeships, increasing idleness and susceptibility to risky coping strategies. Elderly individuals and PADs face significant challenges accessing water and food due to mobility limitations.

Floods in Mnarani disrupt mobility, damage homes, markets, and feeder roads, and compromise WASH infrastructure. Women are heavily affected as they manage contaminated water sources, care for sick or displaced family members, salvage household items, and maintain emergency food supplies. Flood-related market closures significantly reduce women's income from informal trading. Men face reduced earnings from fishing, transport, and daily wage labour. Youth experience disrupted schooling and diminished income opportunities. Elderly persons and PADs face evacuation difficulties and heightened risk of neglect in crowded shelters.

Pests and crop/livestock diseases suppress productivity and affect food security in Mnarani Ward. Women, who dominate subsistence farming and poultry-keeping, experience reduced

yields, while limited access to pest-control inputs, certified seeds, and extension services restricts adaptive capacity. Men face productivity declines in fishing and smallholder farming. Youth lose agricultural casual labour, and elderly persons and PADs experience heightened food insecurity.

Environmental degradation in Mnarani Ward has sharply worsened due to uncontrolled mining and quarrying activities by Mombasa Cement and other operators, coupled with extensive tree cutting for charcoal and mining clearance around Mwangea Hills. These activities have caused severe soil erosion, depleted groundwater, destroyed vegetation cover, and destabilised ecosystems. Women walk longer distances for firewood, water, medicinal plants, and other non-timber forest products, intensifying exhaustion and reducing time for childcare and income generation. Men face declining agricultural and fishing productivity as degraded landscapes reduce streamflow, damage fish breeding areas, and worsen microclimate conditions. Youth lose ecological enterprise opportunities such as beekeeping, tourism, and seedling production. Elderly persons and PADs become increasingly dependent due to limited mobility and reduced access to essential natural resources.

Tree cutting for mineral extraction has reduced natural windbreaks in Mnarani, exposing the ward to more intense strong winds that damage homes, uproot crops, and destroy small-scale infrastructure. Women face heightened caregiving burdens and must secure household assets during storms. Men, especially fishermen, face increased risks of capsizing and gear damage. Youth face injury risks from falling structures, while elderly persons and PADs struggle to access safe shelter, increasing vulnerability.

Coral bleaching is a significant hazard along the Mnarani, Takaungu Creek, where rising sea temperatures have resulted in severe coral mortality. The decline of coral reefs disrupts marine food chains, reduces fish stocks, weakens shoreline protection, and undermines tourism potential. Women involved in gleaning, fish processing, and coastal vending face reduced income and food insecurity due to diminished nearshore resources. Men in artisanal fishing spend more on fuel and take greater safety risks as they venture farther offshore. Youth lose employment in snorkeling, boat operations, and tour guiding. Elderly individuals and PADs experience heightened dependency as households lose marine-based protein sources and income.

Gender-based violence (GBV) further compounds these gendered vulnerabilities, functioning both as a climate-induced risk and as a structural barrier to resilience in Mnarani Ward. Emerging global evidence demonstrates a strong association between climate stressors and rising GBV incidence, with every 1°C increase in temperature linked to a 4.7% rise in intimate partner violence (UN Spotlight Initiative, 2024). Heatwaves have been associated with a 28% increase in femicide, while climate-driven displacement heightens risks of child marriage, sexual exploitation, and trafficking (UN Spotlight Initiative, 2024). In Kenya, rising temperatures and socioeconomic stressors have coincided with escalating femicides with over 930 cases documented between 2016 and 2024, highlighting how climate stress intersects with entrenched gender inequalities (Kenya Femicide Data Study, 2024). These dynamics are most acute in fragile, resource-dependent communities such as Mnarani, where drought, floods, and livelihood loss amplify household tensions, reduce economic security, and disproportionately expose women and girls to physical, psychological, and sexual violence. GBV undermines climate resilience by restricting women's mobility, decision-making autonomy, access to productive assets, and participation in ward-level adaptation processes (UN Women, 2022). As such, GBV must be integrated into anticipatory climate action, beyond response and preparedness, through early warning systems, forecast-based triggers, community surveillance, and pre-positioned support services coordinated with mandated county structures such as the County Gender Department, Social Services, child protection units, and GBV coordination

mechanisms that programmes like COSME have actively engaged. Embedding GBV mitigation within resilience planning ensures that climate adaptation measures safeguard rights, safety, and justice, which are foundational to sustainable and inclusive resilience building.

Table 3: Gender/Age Vulnerability Analysis for Climate Hazards

Women	Men	Youth	Elderly	PADs
Drought				
• Heavy burden of water and firewood collection as sources dry up • Walk longer distances (up to 5 km) → fatigue, dehydration, security risks • Reduced time for childcare and income generation • Psychological stress due to food shortages	• Loss of income from reduced crop productivity and livestock deaths • High stress from inability to provide for households • Reduced availability of casual labour	• Loss of casual labour opportunities • Increased idleness → higher exposure to drugs, crime & unsafe coping • Early marriages and pregnancy risks for girls	• Mobility challenges accessing water and food • High susceptibility to dehydration and heat stress • Increased reliance on caregivers	• High dependency for water, food, and healthcare access • Mobility constraints limit ability to queue for relief supplies • Increased risk of neglect during prolonged shortages
Floods				
• Longer distances to access firewood, water & NTFPs due to land degradation • Increased workload and strain • Reduced time for economic activities • Elevated psychosocial stress	• Loss of fishing, transport, and casual labour opportunities • Psychological stress from inability to provide for families • Damage to tools and assets disrupts livelihoods	• School closures disrupt learning and training • Loss of casual labour -increased idleness, crime, and drug use • Girls particularly vulnerable to early marriage and pregnancy	• High susceptibility to waterborne diseases and cold exposure • Limited mobility restricts evacuation • Greater dependence on others for shelter, food, and medicine	• Severe mobility barriers during evacuation and accessing aid • High risk of injury and neglect in shelters • Limited access to emergency services

Women	Men	Youth	Elderly	PADs
Pests and diseases				
• High exposure due to their role in managing subsistence farms and small livestock • Reduced food security and income from harvest losses • Limited access to pest-control technologies and extension services	• Income loss from crop and livestock declines • Increased stress due to reduced labour demand and food shortages • Higher pressure to diversify livelihoods	• Loss of agricultural casual labour opportunities • Increased unemployment and financial vulnerability • Greater exposure to negative coping behaviours	• Increased dependence for food and care during outbreaks • Reduced capacity to protect crops and livestock • Nutritional deficits worsen chronic illnesses	• High dependency on family for food and adaptive support • Limited access to adapted technologies and extension services • Increased vulnerability from household food shortages
Environmental Degradation				
• Reduced access to firewood, mangroves, herbs, and NTFPs • Walk longer distances (5+ km) for resources leading to exhaustion • Less time for childcare and economic activities	• Declining fish stocks increase offshore travel distance and fuel costs • Higher fishing risks • Reduced forest resource availability increases livelihood strain	• Fewer forest-based livelihood options (beekeeping, nurseries, eco-tourism) • Increased involvement in illegal extraction when opportunities shrink • Rising idleness and economic vulnerability	• Mobility challenges increase dependency during resource scarcity • Physical strain from long distances to collect resources • Heightened vulnerability to malnutrition	• Limited access to degraded resource areas • Exclusion from resource-based livelihood opportunities • Increased reliance on caregivers
Strong winds				

Women	Men	Youth	Elderly	PADs
• Increased caregiving demand after home damage • Higher risk securing household assets during storms	• High danger at sea and gear damage • Increased risk of capsizing during strong winds	• High injury risk from collapsing structures or debris • Limited ability to seek safe shelter	• Difficulty accessing safe shelter quickly • High injury and exposure risk	• Severe challenges evacuating safely • High exposure to falling debris and structural collapse
Coral bleaching				
• Decline in tourism affects women in micro-enterprises (kiosks, vending) • Loss of nearshore ecosystems reduces access to marine resources	• Declining fish stocks reduce artisanal fishing catch • Increased fuel and operational costs as fishing shifts offshore • Higher safety risks at sea	• Reduced employment in snorkelling, tour guiding, and boat operations • Lower household income increases vulnerability to risky coping strategies • Decline in coastal tourism limits livelihood pathways	• Reduced access to fish lowers food security • Limited mobility restricts ability to shift livelihood practices • Higher dependency on family support	• Lost opportunities in tourism and fisheries-based livelihoods • Increased dependency as households face reduced marine resources • Higher risk of exclusion from alternative livelihood training

4.1 Children's Vulnerability to Climate Hazards

Children in Mnarani Ward face heightened vulnerability to climate hazards due to their developmental needs, high dependency on caregivers, and limited physical capacity to cope with environmental stress. The recurrent droughts, seasonal floods, widespread crop and livestock diseases and collectively undermine children's health, safety, nutrition, learning, and emotional well-being.

During droughts, children are among the first to experience nutritional deficits, especially in households dependent on subsistence farming and livestock. Crop failure and livestock mortality reduce access to protein and micronutrient-rich foods, increasing the risk of stunting, acute malnutrition, anaemia, and dehydration. Water scarcity often forces families to rely on unsafe sources, exposing children to diarrhoea, typhoid, and skin infections. In many homesteads, water collection shifts to children, particularly girls, who walk long distances under high temperatures. This results in fatigue, heat stress, early morning school lateness, and reduced class concentration. Prolonged droughts contribute to increased school absenteeism and, in extreme cases, early involvement of children in household labour to supplement family survival.

Floods pose direct physical risks to children, especially in villages adjacent to seasonal rivers and low-lying farms. Floodwaters destroy homes, disrupt Early Childhood Development (ECD) centres, wash away learning materials, and damage WASH facilities. Contaminated floodwater increases exposure to cholera, dysentery, and respiratory infections. Schools are occasionally used as evacuation shelters, halting learning, straining feeding programmes, and weakening psychosocial stability. Children experience anxiety, disrupted routines, sleep disturbances, and loss of a sense of safety when displaced.

Pests and diseases affecting crops and livestock exacerbate household food shortages, directly impacting children's diets. Reduced meal frequency and diminished dietary diversity weaken immunity, making children more susceptible to infections. Strong winds further threaten child safety through collapsing structures and falling trees.

Across all hazards, poverty magnifies child vulnerability. Reduced household income leads families to adopt harmful coping strategies, including withdrawing children from school, increasing child labour, reducing meal portions, or marrying off girls to ease financial strain. These coping mechanisms undermine children's rights, perpetuate intergenerational vulnerability, and reduce long-term resilience.

Protecting children in Mnarani Ward requires a child-centred adaptation approach that integrates early warning systems for households and schools, strengthened nutrition surveillance, and climate-resilient WASH systems. Upgrading schools with storm-resistant roofing, elevated floors, shaded play areas, and reliable water supply will safeguard learning continuity. Establishing psychosocial support and child-friendly spaces during and after climate shocks will help reduce emotional distress. Expanding social protection measures, such as school feeding programmes and targeted cash transfers, will stabilise child well-being during climate-induced livelihood disruptions. Building the capacity of teachers, community health volunteers, religious leaders, and caregivers is essential to ensure that children remain protected before, during, and after climate hazards.

5 Chapter 4: Adaptation Actions

5.1 Introduction

Building on the findings of the hazard and vulnerability analysis, this chapter outlines the adaptation actions designed to strengthen climate resilience in Mnarani Ward. These actions were developed through participatory consultations that utilized Participatory Climate Vulnerability Assessment (PCVA) tools and impact chain analysis to ensure that proposed measures align with the ward's ecological characteristics and socio-economic context. The interventions are tailored to address both the immediate climatic stresses, such as droughts, floods, pests and diseases, environmental degradation, strong winds and coral bleaching and the underlying structural vulnerabilities related to poverty, ecosystem degradation, and institutional capacity gaps.

The proposed measures encompass short-, medium-, and long-term strategies. Short-term actions include community awareness creation, institutional strengthening, and capacity-building initiatives. Medium- and long-term interventions involve climate-resilient infrastructure development, ecosystem restoration, and the promotion of nature-based and technological solutions. For instance, water harvesting at household levels through tanks to mitigate drought, integration of drought tolerant crops to address food shortage, training and capacity building on livelihoods diversification, as well as tree planting and mangrove rehabilitation and restoration will to reduce flooding, coastal erosion, and storm surges.

The formulation of these adaptation actions integrates local knowledge with scientifically validated best practices drawn from Kenya's National Adaptation Plan (2015–2030), the National Climate Change Action Plan (2023–2027), and the Kilifi County Climate Change Action Plan (2023–2027). Gender equity and social inclusion principles are mainstreamed throughout to ensure that adaptation benefits are distributed equitably. Women, who bear a disproportionate burden during droughts and resource scarcity, are prioritized in livelihood diversification, leadership, and financial empowerment programs. Youth are similarly engaged through technology-based initiatives such as climate information dissemination, training, ecosystem monitoring, and innovation in sustainable resource use.

Currently, there are ongoing community-led projects in Mnarani Ward that complement these adaptation priorities, including tree planting, crab farming, mangrove restoration, tree planting initiatives, and financial schemes and programs (credits) that support climate-resilient enterprises. These initiatives demonstrate the community's commitment to sustainable adaptation and provide a strong foundation upon which the proposed actions will build.

Implementation of the adaptation measures will occur at multiple levels i.e., household, community, ward, county, and national, under the coordination of the Ward Climate Change Planning Committee (WCCPC). Collaborative engagement among local organizations CFAs, CBOs, BMUs, and relevant government institutions will be essential to ensure coherent, inclusive, and sustainable delivery of the adaptation agenda across Mnarani Ward.

5.2 Existing climate-resilient and livelihood-enhancing projects

Mnarani Ward currently benefits from a range of community-driven and partner-supported initiatives aimed at enhancing climate resilience, restoring ecosystems, and diversifying livelihoods (Table 3). These interventions collectively strengthen the ward's adaptive capacity through activities such as mangrove restoration, quarry rehabilitation, tree planting, climate-smart agriculture, and water resource management. Efforts led by local committees, community-based groups, and development partners emphasize sustainable natural resource use, capacity building, and locally led adaptation. The ongoing projects also demonstrate collaboration between communities, local area associations and organizations, and the County

Government of Kilifi in advancing ecosystem restoration and climate-resilient livelihoods across the ward.

Mnarani Ward hosts several ongoing initiatives that collectively strengthen community resilience, enhance livelihoods, and promote ecosystem restoration in line with both Kilifi County and national climate change strategies. Central among these efforts are ecosystem protection and restoration programs implemented by the Mnarani Ward Climate Change Committee (WCCC) with similar activities undertaken by Plan International. Activities include the establishment and maintenance of community tree nurseries, catchment rehabilitation, and awareness creation on climate change and mitigation, which contribute to soil conservation, biodiversity recovery, and increased vegetation cover across the Ward.

Coastal and marine ecosystem management remains a focal area, with Plan International and the Mtikamu Community Forest Association (CFA)(Takaungu) implementing mangrove restoration and conservation initiatives within the ward. These include mangrove nursery establishment, replanting of degraded coastal areas, and community sensitization on sustainable ecosystem use. Such interventions align with the National Adaptation Plan (2015–2030), National Climate Change Action Plan (2023–2027), and the National Mangrove Ecosystem Management Plan (2017–2027), contributing to carbon sequestration and shoreline stabilization within Kilifi's broader coastal resilience agenda. Environmental restoration efforts led by Mombasa Cement focus on quarry rehabilitation, where tree planting is undertaken in exhausted quarry sites to restore ecological function and landscape integrity. These initiatives complement county ecosystem restoration priorities under the Kilifi County Integrated Development Plan (CIDP 2023–2027) and support national land rehabilitation measures.

At the community level, the Financing Locally-Led Climate Action (FLLoCA) Project, supported by the World Bank, has enhanced local adaptive capacity through the expansion of water pans and support for community-led adaptation projects. These interventions strengthen water availability, improve drought preparedness, and empower local institutions to manage climate risks more effectively. Cast International is facilitating training and capacity-building programs targeting fisherfolk to diversify income sources and adopt sustainable fishing practices. This helps reduce pressure on marine resources while improving the economic resilience of coastal households. Compassion International complements these actions through education empowerment programs, offering scholarships to vulnerable schoolchildren. This strengthens social resilience by improving access to education and reducing vulnerability linked to poverty, aligning with CIDP 2023–2027 goals for human capital development.

At the same time, the County Government of Kilifi continues to enhance agricultural resilience through extension services that promote climate-smart practices. These efforts reinforce local food security and align with the National Agriculture Sector Climate Change Adaptation Plan objectives. Further, Kenya Forest Service (KFS) and World Vision Kenya are advancing tree planting and forest restoration activities across degraded areas of Mnarani. These initiatives are helping rebuild vegetative cover, conserve critical areas, and promote sustainable natural resource management.

Together, these initiatives illustrate strong coordination among implementing partners in advancing ecosystem resilience, community empowerment, and sustainable livelihoods in Mnarani Ward. They embody the principles of locally led adaptation, ecosystem-based management, and policy alignment, ensuring that community-level actions meaningfully contribute to both county and national climate change priorities.

Table 4: Existing climate-resilient and livelihood-enhancing projects in Mnarani Ward

Project / Activity	Implementing Partner(s)	Location / Target Villages	Adaptation Theme / Focus Area	Key Contributions / Linkages to Existing Strategies
Conservation and Sustainable Management of Coastal and Marine Ecosystems (COSME)	Plan International	Mkwajuni, Vuma, Takaungu, Vjiweni, Midzi mitsano, Timboni	Supporting mangrove nursery establishment Village Savings and Loan Associations (VSLAs), table banking, and gender justice initiatives to empower women economically	Supports afforestation and restoration; aligns with Kilifi County Climate Change Action Plan (2023–2027) on coastal ecosystem restoration; consistent with National Adaptation Plan (2015–2030) and NCCAP (2023–2027) targets on mangrove protection, community-based adaptation, and women's empowerment.
Ecosystem protection and restoration	Mnarani Ward Climate Change Committee (WCCC)	Mnarani ward	Awareness on climate change and mitigation Linking community with climate change related development partners	Supports community engagement and awareness creation under the Kilifi County Climate Change Action Plan (2023–2027); aligns with NAP (2015–2030) objectives for local institutional strengthening and capacity building; consistent with the Climate Change Act (2023) on stakeholder participation.
Environmental restoration	Mombasa Cement	Mkwajuni, Vuma, Takaungu	Planting trees in abandoned quarries	Quarry restoration after exhaustion of mine life; Promotes land rehabilitation consistent with Forest Conservation and Management Act (2016); supports CIDP environmental rehabilitation programs; aligns with NCCAP (2023–2027) measures on land restoration and Kilifi County Climate Change Action Plan targets for ecosystem rehabilitation.
Coastal Ecosystem Restoration	Mtakimau Community Forest Association (CFA)	Takaungu	Mangrove restoration and awareness on climate change and mitigation	Contributes to national mangrove restoration goals under Mangrove Ecosystem Management Plan (2017–2027); supports Kenya Forest Service and Kilifi County Climate Change Action Plan ecosystem management goals; aligns with the Forest Conservation and Management Act (2016).

Project / Activity	Implementing Partner(s)	Location / Target Villages	Adaptation Theme / Focus Area	Key Contributions / Linkages to Existing Strategies
Training and capacity building	Cast International		Empowerment fisher folk groups	Builds local technical capacity and integrates gender-responsive livelihood models; supports ongoing community adaptation programs.
Financing Locally-Led Climate Action (FLLoCA) project	World Bank		Expansion of Benes water pan; Empowers communities to plan fund and implement their own adaptation and resilience projects	Enhancing water security and enhance community resilience to climate change
Education empowerment	Compassion	Vuma	Scholarship provision to school children	Promotes education development, nutrition, and health support; strengthens household resilience
Training and capacity building	County Government of Kilifi	Mnarani ward	Agricultural extension services	Promotes climate smart agriculture and food security
Forest conservation and protection initiatives	Kenya Forest Service	Vuma, Timboni, Takaungu	Tree planting	Supports National Forest Programme; aligns with national targets on afforestation and ecosystem restoration as well as strengthen CFAs capacities
Restore Africa Project	World Vision Kenya	Midzi mitsano, Takaungu	Promotes tree planting and forest restoration to enhance ecosystem resilience	Aligns with Kilifi County Climate Change Action Plan (2023–2027) on restoration objectives as well as compliment national reforestation targets

5.3 Adaptation actions

Adaptation actions in Mnarani Ward are designed as an integrated and gender-responsive climate resilience framework that protects livelihoods, strengthens ecosystems, and safeguards the well-being of all population groups across the six villages; Midzimitsano, Mkwajuni, Takaungu Timboni, Vijiweni and Vuma (Table 5). These actions respond to six priority climate hazards; drought, floods, pests and diseases, strong winds, coral bleaching, and environmental degradation and are anchored in the Kenya Climate Change Act (2023), the National Adaptation Plan (2015–2030), the Kilifi County Climate Change Act (2021), and national frameworks on gender equality, child protection, and sexual and reproductive health and rights (SRHR). Implementation follows a phased approach comprising short-term interventions (≤1 year), medium-term measures (≤3 years), and long-term interventions (>3 years), all anchored in gender equity, social inclusion, SRHR continuity, and community ownership to ensure climate benefits are distributed fairly and sustainably.

The first thematic area emphasizes gender-responsive water security, recognizing that women, girls, and PADs bear a disproportionate burden of water collection and increased domestic care demands during droughts. Short-term actions focus on expanding household rainwater harvesting through the installation of gutters and storage tanks across Midzimitsano, Mkwajuni, Takaungu Timboni, Vijiweni, Vuma, alongside enhancing school water storage with high-capacity tanks to safeguard learning continuity for girls. Medium-term priorities include installing solar-powered pumps on existing wells to reduce the physical strain and long queuing times experienced by women and children, as well as constructing water pans to support domestic use, livestock, and smallholder production. In the long term, resilience will be strengthened through drilling and solarizing community boreholes to ensure a reliable, year-round water supply and reduce the gendered burden of water collection.

The second thematic area focuses on strengthening gender-responsive food and livelihood security, recognizing that women and youth farmers face structural constraints in land access, farm labor, extension services and climate information. In the short term, proposed actions involve training farmers, especially women and youth, on resilience design techniques (from homestead to watershed level) such as soil fertility enhancement, zai pits and labour-efficient farm layouts that reduce production costs and increase adaptive capacity. Additional proposals include training on climate-smart agriculture practices, drought-tolerant crop varieties and fodder conservation methods to stabilize food and livestock systems, alongside sensitizing women and youth groups on crop and livestock insurance products to cushion them from climate-related shocks. Medium-term priorities include supporting selected women and youth lead farmers to establish demonstration plots on CSA and Resilience Design, strengthening extension support, improving access to certified seed and fertilizer, and establishing fodder seed banks and managed pastures that could reduce seasonal feed shortages. Long-term adaptation points toward institutionalizing continuous agricultural learning through the establishment of a Farmer Field School and improving market competitiveness by aggregating produce from women's and youth groups for value addition and enterprise development.

In addition, the thematic area highlights the key role of women, youth, and PADs in natural resource stewardship and aims to strengthen ecosystem restoration for better livelihoods, biodiversity, and shoreline protection. Short-term priorities include training communities to establish tree nurseries, supporting mangrove restoration, promoting organic briquette production, expanding the use of energy-saving jikos and kilns, and improving waste management through composting, sorting, upcycling, and installation of collection bins. Medium-term actions focus on strengthening existing mangrove nurseries and supporting community forest associations to implement their management plans. In the long term, the area envisions integrated mangrove and coral reef restoration to enhance shoreline stability and

ecological resilience, alongside land restoration measures such as terraces and gabions to reduce soil erosion, ensuring that restoration efforts remain inclusive and climate-responsive.

The third thematic area focuses on diversifying climate-resilient livelihoods for women, youth, and PADs. Short-term actions include training communities in alternative livelihoods such as beekeeping, poultry farming, aquaculture, mariculture, ecotourism, and strengthening existing basketry and mat-weaving enterprises through better market linkages. Medium-term efforts involve providing essential equipment such as modern beehives, incubators, fish cages, nets, boats, and feed pellet mills, to reinforce these enterprises and enhance income stability. In the long term, the area aims to support women and youth to establish a mangrove blue carbon project aligned with Plan Vivo and national carbon market regulations, while strengthening VSLA governance and promoting investment of surplus savings in reputable schemes such as green bonds to boost financial resilience.

The fourth thematic area addresses the vulnerability of households, particularly women-headed households, children, elderly persons and PADs, to floods, strong winds, and unsafe settlements. Short-term adaptation actions include training community emergency response teams, undertaking tree planting to reduce wind damage, and designating safe disaster assembly areas and temporary camps to support inclusive evacuation and protection of vulnerable groups. Youth engagement through food-for-work initiatives is proposed to restore drainage channels, clean culverts, construct gabions and plant trees to improve stormwater flow and reduce erosion risks during heavy rains. Medium-term priorities focus on operationalizing multi-hazard early warning systems for floods, drought, strong winds, pests and diseases, as well as improving road drainage and installing culverts to channel floodwater safely and reduce household exposure to hazards. Long-term actions emphasize strengthening community preparedness by equipping emergency response teams with PPEs and first-aid kits and enhancing access to essential services through the construction of health facilities that support disaster readiness and reduce risk for vulnerable populations.

The final thematic area mainstreams gender equality, PAD inclusion, youth empowerment and SRHR continuity across all adaptation actions. Short-term proposals include designing GBV anticipatory actions for drought and floods, assessing existing early warning systems for their capacity to communicate risks to women, girls and PADs, and strengthening awareness on rights, helplines and GBV referral pathways through established structures such as VSLAs. Youth-focused boot camps may address drugs, antisocial behaviour and reproductive health, while PADs could be linked with organizations providing mobility aids, specialized training and entrepreneurship support. Medium-term actions aim to strengthen gender-responsive governance by increasing the participation of women, youth and PADs in ward-level committees and climate decision-making, while equipping local institutions with first-aid kits, rape kits and transport mechanisms to improve GBV response outcomes. In the long term, the ward may consider the establishment of a GBV recovery centre and a youth drug rehabilitation facility at county level, complemented by expanded school feeding programmes, scholarships and vaccination drives to protect the wellbeing of children and adolescents during climate-induced shocks.

Table 5: Mnarani Ward Adaptation actions

Adaptation actions			Potential implementing partners	Budget (KES)
Short-term	*Medium*	*Longterm*		
Thematic area 1: Water Security and Management				
Rainwater harvesting through Installation of gutters on house roofs and water tanks at household level in Midzimitsano, Mkwajuni, Takaungu Timboni, Vijiweni, Vuma villages	Rainwater harvesting through installation of gutters and high-capacity water tanks (10m^3) in schools– one school in each village to enhance water access by women and children	Drilling and solarizing boreholes in Midzimitsano, Takaungu Timboni and Vuma villages-one in each village	Kilifi County Government-FLLoCA, NGOs and Private Sector	50,000/HH, 400,000/School, 5,000,000/borehole
	Installation of solar powered water pumps on existing wells to reduce women workload (turn-around time) in Takaungu, Vijiweni and Vuma villages	Construction of water pans in Midzimitsano, Mkwajuni, Takaungu, Timboni, Vijiweni and Vuma villages		70,000/ water pump, 3,000,000/water pan
Sensitizing women and youth on water treatment in Midzimitsano, Mkwajuni, Takaungu, Timboni, Vijiweni, Vuma villages			KIMAWASCO, Kilifi County Government, CHPs, NGOs and Private Sector	20,000 per village baraza
Thematic area 2: Food security, ecosystem restoration and livelihoods				
Training and supporting women and youth on Resilience Design techniques from homestead to watershed level (Soil fertility improvement, water conservation/retention (e.g. Zai Pits), farm design - to maximize on time and use of local inputs to reduce farming costs)	Supporting select women and youth lead farmers to fully implement resilient design techniques including climate smart agriculture to serve as demonstration farms for continuing learning by other farmers to enhance sustainability of the interventions	Establishing a Farmer Field School in the Ward	Kilifi County Government, KARLO, NGOs and Private Sector	Farmer Field Schoos (FFFs) 10,000 per farmer for a 12-month cycle
Training women and youth on Climate Smart Agriculture (CSA) (drought tolerant crops such as katumani maize cassava, sunflower, groundnuts, cashewnuts, sorghum), silage/hay making to address livestock feed challenges during	Provision of farm inputs (certified seed varieties for drought tolerant crops, organic and inorganic fertilizers, pesticides) and extension services in Midzimitsano, Mkwajuni, Takaungu Timboni, Vijiweni, Vuma villages to trained women and youth groups	Aggregating women and youth farmers' produce to enhance farm produce marketability and enhance value addition and agribusiness opportunities		Production costs: Maize 55,000 per acre and Cassava 40,000 per acre 10,000 for livestock farmers per year or crop cycle (Free if it is

Adaptation actions			Potential implementing partners	Budget (KES)
Short-term	*Medium*	*Longterm*		
drought in Midzimitsano, Mkwajuni, Takaungu Timboni, Vijiweni, Vuma villages		Training women, youth and PADs on cassava value addition Midzimitsano, Mkwajuni, Takaungu Timboni, Vijiweni, Vuma villages		provided by the government)
Sensitize and facilitate access to information on crop and livestock insurance (index-based drought coverage) targeting women and youth groups involved in CSA	Establish livestock fodder seed banks, pastures, and introduce drought-tolerant livestock breeds e.g., Gala goats/Dairy cows for women and youth groups in Midzimitsano, Takaungu Timboni and Vuma villages			Village barazas at 20,000, 55,000 per acre for pasture, 15,000 per Gala goat, 100,000 per dairy cow
Facilitate/train women and youth to establish tree nurseries of priority species (Annex 1) for on-farm tree production (Agroforestry, woodlots and boundary planting) in Midzimitsano, Mkwajuni, Takaungu Timboni, Vijiweni, Vuma villages	Support MTAKIMAU CFA to implement its PFMP programs		Kilifi County Government, KFS, KEFRI, CFAs, NGOs, Private Sector	5,000 per day per trainee (60 pax from the six villages) 75,000 per nursery CFA support based on PFMP budget lines and priorities
Sensitization and training women and youth on mangrove restoration (nursery establishment, transplanting, tending and monitoring) in Takaungu Creek	Assessment and provision of technical support to the existing mangrove nurseries in Takaungu Creek to ensure quality and sustainable seedlings supply for restoration interventions	Implement an integrated mangrove and coral reef restoration programme in Takaungu Creek and Timboni village to enhance shoreline protection and ecosystem resilience respectively	KFS, KEFRI, Kilifi County Government, CFAs, NGOs, Private Sector	5,000 per day per trainee (60 pax from the six villages) 75,000 per nursery 450,000/acre mangrove restoration, 3,000,000/acre coral reef restoration
Training women, youth and PADs on making organic briquettes (coconut husks, cassava peels) in Midzimitsano,	Provision of energy saving jikos and kiln in Midzimitsano, Mkwajuni, Takaungu Timboni, Vijiweni, Vuma villages			5,000 per day per trainee (60 pax from the six villages)

Adaptation actions			Potential implementing partners	Budget (KES)
Short-term	*Medium*	*Longterm*		
Mkwajuni, Takaungu Timboni, Vijiweni, Vuma villages				3,000 energy saving jiko & 7,000 per Kiln
Sensitizing women and youth on proper waste management practices (compositing organic waste, sorting, handling, disposal, upcycling) in Midzimitsano, Mkwajuni, Takaungu Timboni, Vijiweni, Vuma villages	Provision of central waste collection bins in shopping centres and linkage with waste collectors in Midzimitsano, Mkwajuni, Takaungu Timboni, Vijiweni, Vuma villages	Construct terraces and gabions to prevent soil erosion and manage runoff in Vijiweni village	NEMA, WRA, Kilifi County Government, NGOs, Private Sector	20,000 per village barazas 100,000 for waste collection bin BOQ for terraces and gabion construction
Thematic area 3: Diversified Livelihoods and Social Protection				
Training women, youth and PADs on alternative livelihoods such as beekeeping, poultry farming in Midzimitsano, Mkwajuni, Takaungu Timboni, Vijiweni, Vuma villages; ecotourism (mangrove boardwalk) and mariculture (milk fish, mud crab) in Takaungu Village ; and aquaculture (catfish, tilapia) in Midzimitsano village	Provision of alternative livelihoods establishment materials/Equipment; modern beehives, incubators (Midzimitsano, Mkwajuni, Takaungu Timboni, Vijiweni, Vuma villages), boards, fingerlings, cages, nets, boat, pond, and fish feed pellet mill (Midzimitsano and Takaungu villages)		KeFS, KMFRI Kilifi County Government, BMUs, NGOs, Private Sector	5,000 per day per trainee (60 pax from the six villages) 7,000-Modern beehive, 100,000-Incubator, 1,000,000 boardwalk, 15 per fingerling, 1,000,000 cage, 30,000 net, 2,000,000 boat, 100,000 fish feed pellet mill and 1,000,000 pond
Promote basketry and mat weaving (already existing) through providing market linkage initiatives in Midzimitsano, Mkwajuni, Takaungu Timboni, Vijiweni, Vuma villages (exhibition)	Supporting women and youth to initiate and register a mangrove blue carbon project (Plan Vivo Systems and Standards) similar to Vanga blue carbon projects and consistent with the Climate Change (Carbon Markets) Regulations, 2024		NEMA, KEFRI, KFS, Kilifi County Government, CFAs, NGOs, Private Sector	200,000-exhibition 10,000 initial application fee, 100,000 project design document application fee, 1,000,000 feasibility study

Adaptation actions			Potential implementing partners	Budget (KES)
Short-term	*Medium*	*Longterm*		
Train VSLAs on Governance and alternative regenerative investment opportunities such as green bonds and money market funds to diversify interest income from surplus savings	Promote VSLAs excess savings investments in reputable interest earning schemes such as the green bond by Safaricom		County Government of Kilifi, NGOs, Private Sector	5,000 per person per day for training 100,000 per VSLA
Thematic area 4: Community infrastructure and disaster management				
Train selected community emergency response teams in Midzimitsano, Timboni and Vuma villages	Design and implement a food for work programme during drought targeting the youth to restore drainage channels, clean culverts, construct gabions and plant trees for erosion and flood control	Improve road drainage and install culverts to channel floodwater in Timboni village	Kilifi County Government, KERRA, NGOs and Private Sector	BOQs required based on numbers and reputable training institutions
Undertake tree planting to address strong winds in Vuma village Designate flood and wind disaster assembly areas and temporary camps for vulnerable communities in Midzimitsano, Timboni and Vuma villages	Develop and operationalize Early Warning Systems (EWS) for floods, drought, strong winds, pests and diseases (forecasts via chiefs, barazas, media) in Midzimitsano, Takaungu Timboni and Vuma villages	Equip community emergency response teams (PPEs, first aid kits) in Midzimitsano, Takaungu Timboni and Vuma villages Construct a health facility in Vijiweni and Vuma villages	KMD, NDMA Kilifi County Government, Ministry of Interior and National Administration Kenya Red Cross, NGOs and Private Sector	5,000 per day per trainee 70,000 per village for EWS 200,000 disaster assembly area and temporary camp BOQ for equipment and health facility
Thematic area 5: Gender and social inclusion to enhance adaptation action				
Design targeted GBV anticipatory actions for drought, floods and strong winds climate hazards	Develop and implement a Ward GBV anticipatory action plan in collaboration with relevant partners	Establish GBV recovery center (this could be at the county level)	National Government, Kilifi County Government and NGOs, National Government, Kilifi County Government, National Police Service, NGOs, GVRC, COVAW, HAK	1,000,000 for consultancy services Construction of GBV recovery center subject to a detailed Bill of Quantities

Adaptation actions			Potential implementing partners	Budget (KES)
Short-term	*Medium*	*Longterm*		
Assess the existing early warning systems to produce and transmit reliable and timely information and forecasts for the implementation of GBV anticipatory actions ahead of climate hazards	Strengthen gender-responsive governance-inclusion of women, youth and PADs through capacity building in Midzimitsano, Mkwajuni, Takaungu Timboni, Vijiweni, Vuma villages		Kilifi County Government and NGOs	5,000 per person per day for training
Use of existing women associations such as VSLAs to train women and girls on their rights and GBV response strategies, helplines and recovery centers for women and girls as well as the role of men in peaceful conflict resolution	Procure/provide GBV response mechanisms such as transfer vehicles to health facilities and specialized first aid and rape kits to treat and preserve evidence for prosecution respectively		Kilifi County Government, Civil Society, NGOs, Private Sector	5,000 per person per day for training BOQ for GBV response mechanisms
Organize and hold youth boot camps on guidance and counselling focusing on drugs, antisocial behavior and reproductive health	Promote vaccination of children and youth against diseases such as cholera which could arise and spread on account of climate hazards Support school feeding programmes and scholarships	Construct drug rehabilitation centre targeting the youth (this can be done at the county level)	National Government, NACADA, Kilifi County Government, NGOs, Private Sector	100,000- Boot camp, Vaccination at local medical facilities, BOQ required for drug rehabilitation centre School feeding and scholarships needs assessment
Facilitate PADs to link with organizations that can provide infrastructural assistance such as wheelchairs, specialized training and entrepreneur skills			Kilifi County Government, NGOs, Private Sector	50,000 for meetings

5.4 Resource mobilization

Implementation of the Ward Adaptation Plan will rely on a diversified mix of financial, human, and technical resources drawn from county, national, donor, and community-based funding streams. The primary sources include county budget allocations, where relevant departments such as Water, Environment and Natural Resources, Agriculture, Livestock and Blue Economy, and Public Works and Infrastructure will integrate adaptation priorities into their Annual Development Plans and sector budgets. Mnarani Ward is projected to receive at least KES 2–3 million annually earmarked for ward-level climate resilience, as required under the Climate Change Act, 2023.Additional financing will be accessed through the Kilifi County Climate Change Fund (CCCF), which channels grants from development partners including the World Bank's FLLoCA programme, supporting locally led adaptation initiatives. At the national level, Mnarani Ward will benefit from funds such as the National Drought Emergency Fund and other climate financing mechanisms including the Kenya Adaptation Fund Programme, coordinated through the National Treasury's Climate Finance Unit and implemented in collaboration with agencies like NDMA and KMD.

Development partners and NGOs, including Plan International Kenya, World Vision Kenya, Cast International and Compassion, will continue to support ecosystem restoration, capacity building, and livelihood diversification projects, while community contributions through in-kind labour, savings groups, and user fees will promote ownership and sustainability. The private sector will be engaged through partnerships for water infrastructure, crop and livestock insurance, and digital early warning systems, complemented by corporate social responsibility initiatives. Overall, the ward expects to mobilize approximately KES 40–45 million over five years from this blended financing framework, ensuring continuity and long-term sustainability of adaptation interventions.

6 Chapter 5: Adaptation Monitoring and Evaluation

6.1 Introduction

Monitoring, Evaluation, and Learning (MEL) constitute a central pillar of Ward Climate Adaptation Plan, ensuring systematic performance tracking, evidence-based decision-making, and adaptive management throughout the implementation cycle. This chapter delineates the mechanisms through which the adaptation action plan will monitor the progress of adaptation interventions (Monitoring), assess their effectiveness, efficiency, and impact (Evaluation), and institutionalize continuous knowledge generation and feedback loops to strengthen future climate resilience planning (Learning).A technically robust MEL framework will be established to provide accountability, transparency, and data-driven insights to both the community and development partners. This framework will facilitate adaptive management, enabling the Ward Climate Change Planning Committee (WCCPC) to iteratively refine strategies, resource allocation, and implementation modalities based on empirical evidence and evolving climatic realities.

Given the protracted and dynamic nature of climate change, the MEL system will emphasize longitudinal tracking of resilience outcomes demonstrating progressive improvements in ecosystem stability, livelihood diversification, and adaptive capacity over time. The approach will employ participatory monitoring methodologies, integrating community-based data collection and local knowledge with scientifically grounded indicators. This dual approach ensures that both qualitative community insights and quantitative performance metrics inform progress assessments. The MEL framework will align with the County Climate Change Monitoring and Evaluation System and the National Climate Change Monitoring, Reporting, and Verification (MRV) Framework, as mandated under the Climate Change Act, 2023. Compliance with these frameworks guarantees methodological consistency, upward data integration, and contribution to national climate reporting obligations.

Ultimately, the MEL system will serve as a dynamic learning mechanism capturing lessons learned, best practices, and innovations from implementation processes. These insights will inform subsequent revisions of the Ward Adaptation Plan, strengthen institutional capacity, and enhance overall climate governance. By embedding participatory, evidence-based, and adaptive MEL practices. This adaptation plan will ensure that resilience-building efforts are measurable, transparent, and continuously optimized to achieve sustainable climate adaptation outcomes.

6.2 The Adaptation Theory of Change

The theory of change is a conceptual model that explains how the planned interventions will lead to the desired climate resilience outcomes. In simplified form:

Inputs/Activities:

- Capacity building and training for farmers and fisherfolk.
- Infrastructure investments (water pans, boreholes, dams, and gabions).
- Distribution of drought-tolerant seeds, livestock breeds, and energy-efficient technologies.
- Environmental conservation and restoration (tree planting, mangroves, afforestation).
- Strengthened early warning, pests and disease control, and disaster response systems.

Outputs:

- Increased adoption of climate-smart agriculture, mari- culture, aquaculture, and alternative livelihoods.
- Improved access to clean water and better resource management.
- Enhanced community awareness of environmental protection and disaster preparedness

- Fewer flood displacements and decline in reported cases of pest infestations.

Outcomes: The short to medium-term changes due to those outputs – e.g., Increased household resilience through adoption of alternative livelihoods and climate-smart farming, strengthened community preparedness through early warning systems, Reduced crop and livestock losses through vaccination, pest control, and resilient species, improved forest and mangrove cover, Reduced wind damage to farms and property through windbreaks

Impact (Goal): The higher-level objective is enhanced climate resilience and adaptive capacity of the community. This means the community can better withstand droughts, floods, and other hazards without suffering devastating losses, and can recover more quickly. It also includes improved livelihoods and food security under a changing climate, and a reduction in climate-induced emergencies over time.

The theory of change for Mnarani can be summarized as: "If the community and institutions in Mnarani receive training, resources, and infrastructure support to adopt sustainable livelihoods, manage natural resources responsibly, and strengthen climate hazard preparedness, then households will build adaptive capacity, reduce vulnerability to climate shocks, and sustain their livelihoods and ecosystems, leading to a climate-resilient, food-secure, and environmentally sustainable ward." This will lead to the ultimate impact of a more climate-resilient community with secure livelihoods and wellbeing despite climate change.

This logical flow will guide M&E. Each level (input -> output -> outcome -> impact) will have specific indicators to check. For instance, training farmers (input) should lead to output of farmers implementing new techniques, which yields outcome of yield stability in drought, contributing to impact of overall food security. If a link is failing

6.3 Performance Indicators for Adaptation

Based on the theory of change, a set of Performance Indicators will be used to measure progress and effectiveness. These indicators will be both quantitative and qualitative, and they align with national adaptation M&E frameworks where possible (Kenya's NAP/NCCAP indicators) as well as community-defined measures of success. The performance indicators are structured at various levels:

Input/output Indicators:

1. WATER SECURITY AND MANAGEMENT
   - Number of household rainwater harvesting systems installed (gutters + tanks).
   - Number of schools equipped with high-capacity rainwater harvesting systems.
   - Number of water pans constructed or desilted.
   - Number of manual and solar-powered pumps installed on existing wells.
   - Number of women, youth, and PADs using solarised/manual hand pumps.
   - Number and percentage of women, youth, and PADs in water management committees.
   - Number of village water storage structures constructed (boreholes, water pans.
   - Amount of budget allocated and utilised for water infrastructure.
   - Number of boreholes drilled, equipped, and solarised.
   - Number of boreholes fully operational and supplying water year-round.

2. FOOD SECURITY, ECOSYTEM RESTORATION AND LIVELIHOODS
   - Number of farmers trained on Resilience Design techniques.
   - Number of farmers trained on CSA practices (land prep, planting, pest control, post-harvest).
   - Acreage under drought-tolerant crops supported with certified seeds.
   - Number of Farmer Field Schools established and functional.
   - Number of demonstration farms established by women and youth lead farmers.
   - Number of farmers receiving inputs (certified seeds, fertilizers, pesticides).
   - Number of households supported with organic manure production and composting.
   - Number of farmers adopting terracing, zai pits, swales, and other soil/water conservation measures.
   - Number of livestock fodder seed banks and pasture plots established.
   - Number of households receiving improved livestock breeds (Gala goats, dairy cows).
   - Number of farmers sensitised on crop/livestock insurance.
   - Number of households supported with grain storage equipment (hermetic bags, grain banks).
   - Number of village pest and disease surveillance teams established and trained.
   - Number of households receiving indigenous/fruit seedlings.
   - Number of community seedling distribution events held.
   - Number of on-farm tree planting sessions conducted.
   - Number of mangrove nurseries supported and maintained.
   - Number of acres of mangrove restored through enrichment planting.
   - Number of community trainings on mangrove protection and sustainable harvesting.
   - Number of NBS windbreaks (Neem, Casuarina) planted across all villages.
   - Number of environmental conservation awareness sessions held.
   - Number of households adopting sustainable charcoal alternatives or clean energy.
   - Budget allocated and utilised for restoration activities.
   -

3. DIVERSIFIED LIVELIHOODS AND SOCIAL PROTECTION
   - Number of women and youth trained in alternative livelihoods (eco-tourism, beekeeping, poultry).
   - Number of modern beehives, incubators, or eco-tourism boats distributed.
   - Number of households supported with livelihood start-up kits.
   - Number of community groups formed for value addition and market linkages.
   - Number of training sessions on business development, value addition, and financial literacy.
   - Number of social protection initiatives implemented (food distribution, cash transfer mobilizations).
   - Number of school feeding programmes supported.
   - Budget allocations for livelihood diversification implemented.

4. COMMUNITY NFRASTRUCTURE AND DISASTER MANAGEMENT
   - Number of culverts reconstructed or improved with proper drainage.
   - Number of youth mobilised in food-for-work programmes for drainage, gabions, and erosion control.
   - Number of villages with functional flood Early Warning Systems (EWS) linked to KMD.

- Number of community dissemination channels established (chief barazas, noticeboards, SMS).
- Number of community emergency response teams trained.
- Number of community emergency response teams equipped (boats, PPE, first-aid kits).
- Number of designated disaster assembly points established and labelled.
- Number of community trainings on resilient building practices conducted.
- Number of households implementing safer roofing techniques (replacing strings with nails).
- Budget allocated and utilised for infrastructure upgrades (roads, culverts, drainage).

5. GENDER AND SOCIAL INCLUSION TO ENHANCE ADAPTATION ACTIONS

- Number of gender-responsive governance trainings conducted.
- Number and percentage of women, youth, and PADs participating in ward committees.
- Number of VSLAs supported with GBV awareness and rights training.
- Number of GBV anticipatory action sessions conducted for drought, floods, and strong winds.
- Number of GBV response kits (first-aid kits, rape kits) distributed to health facilities.
- Number of early warning systems integrating GBV alert messaging.
- Number of PADs supported with assistive devices, rehabilitation linkages, or livelihood equipment.
- Number of youth bootcamps conducted on drugs, behaviour change, and reproductive health.
- Number of awareness campaigns on GBV helplines and safe reporting mechanisms.
- Number of meetings with PAD support organizations facilitated.

Outcome Indicators:

1. WATER SECURITY AND MANAGEMENT

- Reduction in average time spent collecting water (minutes/day), disaggregated by women, youth, and PADs.
- Increase in household water storage capacity (litres/HH).
- Percentage of water facilities functional year-round.
- Increase in number/percentage of women, youth, and PADs with independent access to water sources (boreholes, solarised pumps, kiosks).

- Increased reliability of water supply during drought periods (days/months of service continuity).

2. RFOOD SECURITY, ECOSYTEM RESTORATION AND LIVELIHOODS

- Stability of crop yields over 3–5 seasons among supported farmers.
- Improvement in soil health indicators, including:
- Increase in soil organic matter (%)
- Increase in soil moisture retention capacity
- Improvement in soil nutrient status (N, P, K)
- Improved aggregate stability and soil structure
- Percentage of farmers adopting integrated soil and water conservation (terracing, zai pits, swales).
- Increase in acreage under diversified and climate-resilient farm systems.

- Increase in household-level adoption of post-harvest management technologies (grain banks, hermetic bags).
- Percentage of women, youth, and PADs participating in CSA and Resilience Design training cycles.
- Increase in on-farm and community tree cover (%) over baseline.
- Increase in mangrove seedling survival rates and total acreage restored.
- Reduction in incidents of illegal harvesting, charcoal burning, and mangrove cutting.
- Increase in households adopting windbreak and shelterbelt systems.
- Increased community participation in environmental conservation and sustainable harvesting practices.

3. DIVERSIFIED LIVELIHOODS AND SOCIAL PROTECTION

- Percentage increase in household income from diversified livelihoods (eco-tourism, poultry, beekeeping, cassava value addition).
- Increase in the number of households with multiple income streams across seasons.
- Percentage of women, youth, and PADs accessing new livelihood opportunities.
- Increased financial literacy and business skills among trained groups.
- Increase in households benefiting from school feeding, cash transfers, or food-for-work schemes.

4. COMMUNITY INFRASTRUCTURE AND DISTATER MANAGEMENT

- Reduction in households exposed to flood-prone or erosion hotspots.
- Percentage of villages receiving timely Early Warning System (EWS) alerts.
- Community response rate to EWS alerts (attendance at assembly points, evacuation compliance).
- Reduction in flood-related damages to dwellings and access roads.
- Level of preparedness and readiness of community emergency response teams (measured via drills).
- Increased adoption of wind-resilient building practices in households.

5. GENDER AND SOCIAL INCLUSION TO ENHANCE ADAPTATION ACTIONS

- Increase in the number and proportion of women, youth, and PADs participating in ward-level governance and adaptation committees.
- Reduction in drudgery for women, youth, and PADs due to improved water access and resilient infrastructure.
- Percentage increase in women's and PADs' access to credit, extension services, and productive resources.
- Increase in women's ownership of productive assets (livestock, equipment).
- Improvement in community knowledge and awareness of GBV risk mitigation during climate hazards.
- Increased utilisation of GBV response services (safe shelters, first-aid kits, referral pathways).

Impact (Goal) Indicators:

1. WATER SECURITY AND MANAGEMENT

- Percentage of the ward population with year-round access to safe and reliable water.

- Increase in average litres of water available per person per day across villages.
- Long-term reduction in water scarcity–related vulnerabilities for women, youth, and PADs.
- Sustained decrease in waterborne diseases due to improved water quality and availability.
- Community-wide resilience to prolonged drought through diversified water sources and storage systems.

2. FOOD SECURITY, ECOSYTEM RESTORATION AND LIVELIHOODS

- Sustained increase in crop yields above baseline across 3–5 year periods.
- Long-term reduction in seasonal yield fluctuations linked to droughts, floods, and pest outbreaks.
- Measurable improvements in soil health at landscape level, including:
- Higher soil organic carbon (SOC)
- Increased vegetation cover (NDVI index)
- Reduced erosion and sediment loss
- Enhanced household food security across all seasons.
- Increased biodiversity within integrated farming systems (trees, fodder, pollinator presence).
- Increases in forest cover, on-farm trees, and mangrove extent compared to baseline.
- Restoration of degraded landscapes evidenced by improvements in NDVI and tree density.
- Reduction in illegal harvesting, mangrove cutting, and charcoal burning.
- Enhanced ecosystem services, including shoreline protection, soil stability, and microclimate regulation.
- Long-term improvement in habitat quality for biodiversity and fisheries productivity.

3. DIVERSIFIED LIVELIHOODS AND SOCIAL PROTECTION

- Sustained increases in household incomes from diversified livelihoods (beekeeping, eco-tourism, poultry).
- Improved household resilience to income shocks from climate hazards.
- Decreased poverty levels due to stable and multi-stream livelihoods.
- Increased school attendance and nutritional outcomes for children due to social protection schemes.
- Greater financial stability among women, youth, and PAD households.

4. COMMUNITY INFRASTRUCTURE AND DISTATER MANAGEMENT

- Significant reduction in climate-related damage to homes, roads, and community facilities.
- Communities demonstrating effective disaster preparedness and response during floods and storms.
- Long-term reduction in displacement, injuries, and asset losses during climate hazards.
- Increased structural safety and durability of settlements in strong-wind and flood-prone areas.

5. GENDER AND SOCIAL INCLUSION TO ENHANCE ADAPTATION ACTIONS

- Percentage increase in women's, youth's, and PADs' participation in climate governance and local decision-making structures.
- Strengthened continuity of SRHR, maternal health, and GBV response services during climate hazards.
- Increased leadership roles held by women, youth, and PADs in adaptation implementation.

- Growth in the number of marginalised and vulnerable groups leading or influencing climate adaptation actions.
- Reduction in GBV risks during droughts, floods, and strong winds due to anticipatory action systems.

For each indicator, we will establish baselines (where we are starting from, using historical data or initial surveys) and set targets for 1-year, 3-year, 5-year intervals. For example: baseline of 0 operational boreholes, target 3 by 2025; baseline average maize yield 8 bags/acre in normal year and 2 in drought year, target to achieve 5 in a drought year with adaptation by 2027; baseline women fetching water 5 km, target <1 km by 2024, etc.

Data for these indicators will be obtained from various sources: routine project monitoring (engineer reports for infrastructure, training attendance sheets), surveys and focus groups for outcomes (e.g., seasonal household surveys to gauge water access, yields, or time saved), and existing data (health clinic records for disease incidence, acreage of mangroves restored).

All indicators will be disaggregated by gender where applicable to monitor differential benefits. Also, where feasible, by village to see if some areas lag. Measuring adaptation success is complex because it's about avoided losses and capacity built, but these indicators together provide a robust picture. The Ward Committee will use a simple M&E matrix listing each indicator, baseline, target, data source, frequency of data collection, and responsible person.

6.4 Adaptation Reporting and Learning

Reporting: Mnarani Ward will establish a structured and transparent reporting system to facilitate timely information flow among all stakeholders, including community members, county authorities, and development partners. The Ward Climate Change Planning Committee (WCCPC) will prepare quarterly progress reports detailing implemented activities, achieved outputs, encountered challenges, and corresponding expenditures. These reports will be presented in public barazas to promote accountability and enable community feedback and subsequently submitted to the County Climate Change Unit for integration into county-level monitoring processes. In addition, an annual Ward Climate Adaptation Report will be developed to comprehensively assess progress against planned targets and performance indicators. This annual report will align with the requirements of the Climate Change Act (2023), ensuring that Mnarani Ward's results and lessons contribute to the counties and nation's broader climate change monitoring, reporting, and verification (MRV) systems.

The annual report will highlight case studies or success stories (e.g., "Farmer groups in Vuma doubled their harvest despite the 2025 drought by using new techniques a success attributed to our adaptation training"). It will also not shy away from discussing failures or shortfalls, as these provide learning.

At community level, participatory monitoring meetings will be held. For instance, after each rainy season, community members can gather to discuss amount mangrove seedlings planted. Did the new seeds perform better in this season? Such qualitative feedback will be recorded and included in reports.

The Ward Administrator and WCCPC Chair will also report in county forums (e.g., County Adaptation Planning Committee meetings) on progress, championing their ward and possibly leveraging more support by showcasing results.

Learning and Reflective Review: Adaptation is a long journey with uncertainty, so fostering a culture of learning is vital. Twice during the implementation period (e.g., mid-term in year 3, and end-term in year 5), the ward will conduct an outcome evaluation and learning workshop. This involves bringing together community representatives, implementing partners, and possibly external evaluators to systematically review what outcomes have been achieved and why. Key questions: Which interventions are most effective? Are there unintended impacts? What obstacles are reducing effectiveness (maybe maintenance issues, or cultural barriers to adoption)? For example, maybe the uptake of an improved cookstove program is low – through reflection the group might learn women prefer a different design, or fuel is an issue, leading to adjusting that action. These learning workshops will use tools like community scorecards, ranking of interventions by community benefit, and problem tree analysis for any persistent challenges (similar to the PCVA style but now focusing on project performance). Lessons learned will be documented and used to refine the adaptation plan. The Ward Climate Committee is empowered to tweak or reprioritize actions based on evidence. For instance, if a particular crop variety fails, they will note it and try an alternative next season.

Knowledge Sharing: Mnarani Ward will also engage in horizontal learning i.e., exchanging experiences with other wards or counties. If something worked really well here (say the VSLA approach to fund water tanks), we can host visitors from nearby wards to showcase it, and similarly we can learn from others (maybe a neighbouring ward piloted a solar irrigation scheme that we could replicate). The county climate change unit can facilitate such inter-ward learning exchange visits or meetings.

Feedback to Policy: Through reporting channels, the ward's experiences may also inform higher level policies. For example, if the ward finds that insurance uptake is low because of certain policy constraints, this feedback can go to the national level via NDMA or the National Climate Change Council to refine such programs. The 2023 Act calls for progressive improvement of climate response, our ward's data and stories become part of that national learning loop.

Sustainability of M&E: The Ward Committee will assign an M&E focal person (or sub-committee) to drive these activities. But monitoring won't be just "external evaluation"; it will be integrated into community life e.g., water committees will track water usage and downtime of boreholes as part of their routine, farmer groups will collectively observe and record crop performance. By embedding monitoring tasks into existing structures, we ensure continuity even if specific projects end.

7 References

1. African Union Development Agency–NEPAD. (2020). Climate risk and vulnerability in Africa: Mapping hotspots and adaptation priorities. AUDA–NEPAD.
2. Africa Climate Policy Centre. (2020). Climate change projections for Eastern Africa. United Nations Economic Commission for Africa.
3. Arora-Jonsson, S. (2011). Gender inequality and climate change: Misconceptions and gendered policy responses. The Journal of Peasant Studies, 38(1), 93–109.
4. CHIRPS. (2022). Climate Hazards Group InfraRed Precipitation with Station data. University of California, Santa Barbara. https://www.chc.ucsb.edu/data/chirps
5. County Government of Kilifi (2021). Kilifi County Climate Change Act. County Government of Kilifi.
6. County Government of Kilifi. (2021). County Climate Change Fund (CCCF) Regulations. County Government of Kilifi.
7. County Government of Kilifi. (2023). Kilifi County Climate Change Action Plan (2023–2027). County Government of Kilifi.
8. County Government of Kilifi. (2023). Kilifi County Integrated Development Plan (CIDP III 2023–2027). County Government of Kilifi.
9. County Government of Kilifi. (2020). County Spatial Plan (2021–2030). Kilifi County.
10. COSME – Conservation and Sustainable Management of Coastal and Marine Ecosystems. (2023). Participatory Climate Vulnerability and Capacity Assessment (PCVA) reports for coastal wards. Plan International.
11. Doss, C. R., Kovarik, C., Peterman, A., Quisumbing, A. R., & van den Bold, M. (2015). Gender inequalities in ownership and control of land in Africa. Agricultural Economics, 46(3), 403–422.
12. EMCA. (2015). Environmental Management and Coordination Act (Amendment 2015). National Council for Law Reporting.
13. ERA5-Ag. (2022). Agro-meteorological indicators from ERA5 reanalysis. Copernicus Climate Data Store.
14. Envasses Environmental Consultants Limited. (2025). Participatory Climate Vulnerability and Capacity Assessment (PCVA) reports for Midzimitsano, Mkwajuni, Takaungu Timboni, Vijiweni &Vuma villages villages. Envasses.
15. Food and Agriculture Organization. (2017). Kenya Climate Smart Agriculture Strategy 2017–2026. FAO.
16. Food and Agriculture Organization. (2020). Ecosystem-based adaptation in the agriculture sector: Guidance for practitioners. FAO.
17. Global Resilience Design. (2023). Resilience design handbook: Regenerative water, soil & biodiversity management. https://www.globalresiliencedesign.com
18. Government of Kenya. (2013). Fisheries Policy of Kenya. Ministry of Agriculture, Livestock & Fisheries.
19. Government of Kenya. (2015). National Adaptation Plan (NAP) 2015–2030. Ministry of Environment and Forestry.
20. Government of Kenya. (2018). National Disaster Risk Management Policy. Ministry of Interior.
21. Government of Kenya. (2019). Kenya Population and Housing Census 2019. Kenya National Bureau of Statistics.
22. Government of Kenya. (2020). National Framework for Climate Services (NFCS). Ministry of Environment and Forestry.
23. Government of Kenya. (2023). Climate Change (Amendment) Act, 2023. National Council for Law Reporting.

24. Government of Kenya. (2023). Kenya's Updated Nationally Determined Contribution (NDC 2023). Ministry of Environment, Climate Change and Forestry.
25. IGAD Climate Prediction and Applications Centre. (2022). Eastern Africa climate outlook. ICPAC.
26. Intergovernmental Panel on Climate Change. (2021). Climate change 2021: The physical science basis. Cambridge University Press.
27. Intergovernmental Panel on Climate Change. (2022). Climate change 2022: Impacts, adaptation & vulnerability. Cambridge University Press.
28. KALRO – Kenya Agricultural and Livestock Research Organization. (2022). Drought-tolerant crops & climate-smart agriculture briefs. KALRO.
29. KEMFSED – Kenya Marine Fisheries and Socio-Economic Development Project. (2021). Project progress reports for coastal counties. State Department for Fisheries.
30. Kenya Forest Service. (2022). Forest and Landscape Restoration Implementation Plan (FOLAREP 2022–2027). KFS.
31. Kenya Forest Service. (2023). National Mangrove Ecosystem Management Plan. KFS.
32. Kenya Marine & Fisheries Research Institute. (2021). Marine ecosystem and fisheries status assessments for the South Coast. KMFRI.
33. Kenya Meteorological Department. (2023). Kilifi County climate outlook & localized weather advisories. KMD.
34. KidsCare Kenya. (2022). Disability inclusion and GBV awareness reports for Kilifi County. KidsCare Kenya.
35. Kilifi County Government. (2023). Kilifi County Participatory Climate Risk Assessment (PCRA). Kilifi: County Government of Kilifi
36. KOMAZA. (2021). Smallholder commercial forestry model reports. Komaza Ltd.
37. Landsat. (2023). Landsat surface reflectance products (5, 7, 8 & 9). United States Geological Survey.
38. Ministry of Environment & Forestry. (2020). National Mangrove Restoration Strategy 2020–2030. Government of Kenya.
39. Ministry of Health Kenya. (2021). Clinical management of GBV survivors: National guidelines. Government of Kenya.
40. Ministry of Water & Irrigation. (2019). National Water Master Plan 2030. Government of Kenya.
41. National Drought Management Authority. (2022). Drought early warning bulletins for Kilifi County. NDMA.
42. Paris Agreement. (2015). Paris Agreement under the United Nations Framework Convention on Climate Change. United Nations.
43. Red Cross Kenya. (2021). Community-based disaster risk management manual. Kenya Red Cross Society.
44. Sendai Framework for Disaster Risk Reduction 2015–2030. (2015). United Nations Office for Disaster Risk Reduction.
45. State Department for Fisheries. (2020). Annual fishery catch statistics for the coastal region. Government of Kenya.
46. State Department for Gender. (2019). National policy for the prevention and response to gender-based violence. Government of Kenya.
47. TripleOKLaw LLP. (2023). Overview of the Climate Change (Amendment) Act 2023 – Devolved climate governance & Ward Climate Change Committees. https://www.tripleoklaw.com
48. UN Women. (2022). Gender and climate change: Programming guidance & evidence review. UN Women.

49. UNEP – United Nations Environment Programme. (2018). Nature-based solutions for climate resilience. UNEP.
50. UNICEF. (2021). Children and climate change in Eastern and Southern Africa. UNICEF.
51. United Nations. (2015). Transforming our world: The 2030 Agenda for Sustainable Development (SDGs 2, 5, 6, 13 & 14). UN General Assembly.
52. USAID. (2020). Climate risk profile: Kenya. USAID Climate Change Office.
53. WIOMSA – Western Indian Ocean Marine Science Association. (2022). Western Indian Ocean coastal and marine ecosystem technical reports. WIOMSA.
54. World Bank. (2015). Kenya Coastal Development Project (KCDP) completion report. World Bank.
55. World Bank. (2021). Financing Locally Led Climate Action (FLLoCA) Programme implementation manual. World Bank Group.
56. World Vision Kenya. (2022). Forest landscape restoration and community resilience reports for coastal Kenya. World Vision Kenya.
57. World Vision Kenya. (2023). Climate adaptation and livelihood diversification reports for Kilifi & Kilifi counties. World Vision.
58. WWF Kenya. (2022). Forest Landscape Restoration (FLR) products value chain and market analysis – Loitokitok Sub-County. WWF Kenya.
59. WWF Kenya. (2022). Triple Benefit Project: Nature-based solutions assessment for Kilifi and Lamu. WWF Kenya.

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9 Annexure

1. Priority tree and fruit species

Scientific Name	Common Name
Azadirachta indica	Neem
Afzelia quanzensis	Pod Mahogany / Mkongo
Casuarina equisetifolia	Whistling Pine / Casuarina
Milicia excelsa	African Teak / Iroko / Mvule
Gmelina arborea	Gmelina / White Teak
Tectona grandis	Teak
Grevillea robusta	Silk Oak / Grevillea
Senna spp.	Senna / Cassia
Moringa oleifera	Moringa / Drumstick Tree
Azalea spp.	Azalea
Citrus sinensis	Sweet Orange
Anacardium occidentale	Cashew
Mangifera indica	Mango
Cocos nucifera	Coconut
Carica papaya	Papaya